5G RAN

Service-differentiated Experience Guarantee Feature Parameter Description

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Contents

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for uplink and downlink decoupling of the Unlimited Rate Switch for data rate limitation in flexible experience differentiation. For details, see: • **9.1.1 Data Rate Limitation in Flexible Experience Differentiation** • **9.4.1.1 Data Preparation** • **9.4.1.2 Using MML Commands**	Modified parameter: Added the **UNLIMITED_RATE_SW** option of the **gNBFlexExpDiffCfg**.***EquityLevelAlgoSw*** parameter to the disuse list and replaced it with the **UL_UNLIMITED_RATE_SW** and **DL_UNLIMITED_RATE_SW** options of this parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Changed Flexible Rate Grading from a commercial feature into a trial feature. For details, see **2.1 General Statements**, **2.2 Features in This Document**, and **9.3.1 Licenses**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Incorporated the following change: The constraints of Flexible Rate Grading take effect only in SA scenarios. For details, see Differences and **9.1 Principles**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Deleted descriptions of the Smooth Video Interaction feature because this feature is not recommended for commercial use in the current version.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

  Added descriptions of "frequency selection policy for QCI-based preallocation". For details, see **4.1.1 QCI-based Uplink Preallocation**.

  Revised descriptions of deactivation command examples for QCI-specific PDCCH aggregation level optimization. For details, see **5.4.1.2 Using MML Commands**.

Revised descriptions of the impact relationship between uplink non-consecutive scheduling and "data rate limitation in flexible experience differentiation". For details, see **9.3.2.3 Function Impacts**.

Deleted descriptions of data preparation and MML-based configurations for the uplink non-consecutive scheduling switch in "data rate limitation in flexible experience differentiation" from **9.4.1.1 Data Preparation** and **9.4.1.2 Using MML Commands**.

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 07 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for the Flexible Rate Grading feature. For details, see **9 Flexible Experience Differentiation**.	Added the **gNBFlexExpDiffCfg** MO. Added the **NRDUCellUuPrfmOpt**.*DiffTargetRateCalcPeriod*, **NRDUCellQciBearer**.*EquityLevel*, **NRDUCellPusch**.*UlDiffTargetRateEstFactor*, **NRDUCellUuPrfmOpt**.*DiffTargetRateEstFactor*, and **NRDUCellPusch**.*UlFlexExpDiffExitRateThld* parameters. Modified parameter: Added the **FLEX_EXP_DIFF_SW** and **FLEX_EXP_DIFF_RPT_SW** options to the **NRDUCellFeatureSw**.*QciBasedDiffServExpSw* parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between downlink experience satisfaction guarantee and flexible experience differentiation. For details, see **9.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added support for the QCI-specific CCE aggregation level adaptation function. For details, see **6 QCI-specific CCE Aggregation Level Adaptation**.	Modified parameter: Added the **CCE_AGG_LEVEL_ADAPT_SW** option to the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter. Added the **gNBDUSchParamGroup**.***CceAllocFailRateUpperThld*** and **gNBDUSchParamGroup**.***CceAllocFailRateLowerThld*** parameters.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for the Smooth Video Interaction feature.	Modified parameters: • Added the **VIDEO_INTERACT_EXP_GUAR_SW** option to the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter. • Added the **PUSCH_HARQ_DTX_PDCCH_ADJ_SW**, **UL_MCS_ADJ_ALGO_ADAPT_SW**, and **DIFF_PUSCH_DMRS_FDM_SW** options to the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter. Added the **gNBDUSchParamGroup**.***UlMcsAdjAdaptThld*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added policies for controlling the number of UEs for preallocation. For details, see: • **4.1.1 QCI-based Uplink Preallocation** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Added the **NRDUCellUlsch**.***QciPreschUeNumLimit*** and **NRDUCellUlsch**.***CellPreschUeNumLimit*** parameters. Modified parameter: Added the **PRESCH_UE_NUM_CTRL_POL_SW** option to the **NRDUCellUlSch**.***UlPreallocPolEnhSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for "QCI-specific service optimization for specific UEs". For details, see **4.1.4 QCI-specific Scheduling Based on the Target IBLER**.	Modified parameter: Added the **SPEC_USR_SERVICE_OPT_QCI_SW** option to the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for activation verification for "differentiated gap-assisted measurement" and "explicit scheduling for downlink second retransmission" via QCI-specific delays at the MAC layer in NSA networking. For details, see **5.4.2 Activation Verification**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for configuring the **ANBR Resource Factor** parameters. For details, see **7 MAC CE-based Rate Adjustment for Data Services**.	Added the **NRDUCellDlSch**.***AnbrResFactor*** and **NRDUCellUlSch**.***AnbrResFactor*** parameters.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

  Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-050208	Service-Differentiated Low Latency	**4 Service-differentiated Low Latency**
FOFD-091202	Game-like Service Latency Assurance	**5 Service-differentiated Link Reliability**
FBFD-091100	ANBR	**7 MAC CE-based Rate Adjustment for Data Services**
FOFD-091230	QoS-based CA Policy	**8 QoS-based CA Policy**
FOFD-100210	Flexible Rate Grading (trial)	**9 Flexible Experience Differentiation**

3 Overview

On live networks, there are various services (such as video, cloud gaming, and voice services) with different network requirements. To meet such requirements, the base station can differentiate between services through 5QIs/QCIs and provide service-differentiated experience guarantee functions, as listed in **Table 3-1**.

Table 3-1 Functions in service-differentiated experience guarantee

Function Name	Sub-function Name	Overview	Chapter/Section
Service-differentiated low latency	QCI-based uplink preallocation	The gNodeB allows for differentiated preallocation parameter group configurations among QCI-specific bearers.	**4.1.1 QCI-based Uplink Preallocation**
	QCI-specific SR period configuration	An SR period can be customized for services carried on a QCI-specific bearer.	**4.1.2 QCI-specific SR Period Configuration**
	QCI-specific first-packet size configuration for SR-based scheduling	A first-packet size can be customized for a QCI-specific bearer in SR-based scheduling.	**4.1.3 QCI-specific First-Packet Size Configuration for SR-based Scheduling**
	QCI-specific scheduling based on the target IBLER	Uplink and downlink target IBLERs can be configured for a specific QCI.	**4.1.4 QCI-specific Scheduling Based on the Target IBLER**

Function Name	Sub-function Name	Overview	Chapter/Section
Service-differentiated link reliability	QCI-specific PDCCH aggregation level optimization	For UEs with QCI-specific bearers, the PDCCH SINR and PDCCH BLER are adjusted, and the PDCCH aggregation level is adaptively adjusted based on the PUSCH DTX status, reducing the probability of DCI miss-detection. In this way, the uplink packet loss rate is reduced.	**5.1.1 QCI-specific PDCCH Aggregation Level Optimization**
	QCI-specific uplink MCS selection optimization	The uplink MCS indexes for initial transmissions of UEs with QCI-specific bearers are reduced to decrease the uplink packet loss rate.	**5.1.2 QCI-specific Uplink MCS Selection Optimization**
	QCI-specific downlink MCS selection optimization	For UEs with QCI-specific bearers, the gNodeB decreases the CQI by a fixed value (1, 2, or 3) if HARQ feedback for initial transmission is NACK. This reduces the downlink MCS index, thereby lowering the downlink packet loss rate.	**5.1.3 QCI-specific Downlink MCS Selection Optimization**
	QCI-specific uplink MU pairing differentiation	This function prohibits the UEs with QCI-specific bearers from participating in PUSCH MU-MIMO pairing, which reduces inter-layer interference of pairing.	**5.1.4 QCI-specific Uplink MU Pairing Differentiation**

Function Name	Sub-function Name	Overview	Chapter/Section
	QCI-specific downlink MU pairing assurance	This function lowers the correlation threshold for downlink MU-MIMO pairing between UEs with QCI-specific bearers and other UEs. This makes pairing between them more difficult and mitigates inter-layer interference of pairing.	**5.1.5 QCI-specific Downlink MU Pairing Assurance**
	QCI-specific downlink power control optimization	The PDSCH power spectral density (PSD) is adaptively adjusted based on the remaining power after scheduling, thereby achieving downlink power compensation for UEs with QCI-specific bearers.	**5.1.6 QCI-specific Downlink Power Control Optimization**
	QCI-specific uplink power control optimization	If there is remaining uplink transmit power for UEs with QCI-specific bearers at the cell center or at a medium distance from the cell center, the PUSCH PSD can be increased to improve link reliability.	**5.1.7 QCI-specific Uplink Power Control Optimization**
	Differentiated gap-assisted measurement	This function specifies whether UEs with a specific QCI participate in gap-assisted measurement.	**5.1.8 Differentiated Gap-assisted Measurement**

Function Name	Sub-function Name	Overview	Chapter/Section
	Explicit scheduling for downlink second retransmission	For UEs with QCI-specific bearers, if the base station detects that the first-retransmission result reported by the UEs is NACK and the transport block size (TBS) allowed at the second retransmission is greater than or equal to the TBS allowed at the initial transmission, the base station uses explicit scheduling for retransmission. That is, the base station sends the DCI used for retransmission to the UEs to indicate a specific MCS index. This increases the demodulation success rate of the UEs.	**5.1.9 Explicit Scheduling for Downlink Second Retransmission**
QCI-specific CCE aggregation level adaptation	NA	Whether "QCI-specific PDCCH aggregation level optimization" is required can be determined based on the cell's uplink and downlink CCE allocation failure rates for UEs with QCI-specific bearers.	**6 QCI-specific CCE Aggregation Level Adaptation**
MAC CE-based rate adjustment for data services	N/A	The base station uses MAC CEs to notify a UE of the available uplink or downlink rate in accordance with the air interface quality. Then, the UE can adjust the uplink or downlink bit rate accordingly to match the available uplink or downlink rate.	**7 MAC CE-based Rate Adjustment for Data Services**

Function Name	Sub-function Name	Overview	Chapter/Section
QoS-based CA policy	N/A	In carrier aggregation (CA) scenarios, CA frequency combinations can be differentiated for UEs with specific QCIs.	**8 QoS-based CA Policy**
Flexible experience differentiation	Data rate limitation in flexible experience differentiation	Differentiated and customizable data rates are enabled by (1) setting different privilege levels for users with different QCIs, (2) configuring different target data rate proportions and minimum guaranteed bandwidths for privilege levels, and (3) periodically calculating the uplink or downlink peak user rate.	**9 Flexible Experience Differentiation**
	Data rate reporting in flexible experience differentiation	The base station reports the estimated uplink or downlink data rate for each privilege level to the core network based on the negotiation result between the base station and the core network.	

4 Service-differentiated Low Latency

4.1 Principles

The service-differentiated low latency function is introduced to reduce the delay over the air interface for UEs with QCI-specific bearers. **This function enables differentiated scheduling parameter configurations among QCI-specific bearers. This reduces the processing time required for scheduling, the block error rate (BLER), and the number of retransmissions, thereby decreasing the delay over the air interface.**

The service-differentiated low latency function is controlled by the **SERV_DIFF_LOW_LATENCY_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter. The following table lists the sub-functions of this function.

Sub-function Name	Description	Reference
QCI-based uplink preallocation	The gNodeB allows for differentiated preallocation parameter group configurations among QCI-specific bearers.	**4.1.1 QCI-based Uplink Preallocation**
QCI-specific SR period configuration	An SR period can be customized for services carried on a QCI-specific bearer.	**4.1.2 QCI-specific SR Period Configuration**
QCI-specific first-packet size configuration for SR-based scheduling	A first-packet size can be customized for a QCI-specific bearer in SR-based scheduling.	**4.1.3 QCI-specific First-Packet Size Configuration for SR-based Scheduling**
QCI-specific scheduling based on the target IBLER	Uplink and downlink target IBLERs can be configured for a specific QCI.	**4.1.4 QCI-specific Scheduling Based on the Target IBLER**

NOTE

QCI-based uplink preallocation and QCI-specific scheduling based on the target IBLER take effect only when the **SERV_DIFF_LOW_LATENCY_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter is selected.

4.1.1 QCI-based Uplink Preallocation

Uplink preallocation is a mechanism whereby the gNodeB proactively schedules a UE at a specified interval regardless of whether the UE has sent an SR to the gNodeB. This mechanism avoids the loop that persists from when the UE sends an SRI to when the UE obtains an uplink scheduling grant. Uplink preallocation is classified into uplink basic preallocation and uplink smart preallocation, which differ in how they take effect. For details, see *Uplink Scheduling*.

Given that uplink preallocation takes effect on a per cell basis and cannot distinguish between UEs of different types, QCI-based uplink preallocation is introduced. **This function enables differentiated preallocation parameter group configurations among QCI-specific bearers by the gNodeB, reducing the processing time required for scheduling.** QCI-based uplink preallocation is classified into QCI-based uplink basic preallocation and QCI-based uplink smart preallocation, which differ in how they take effect and cannot be enabled at the same time.

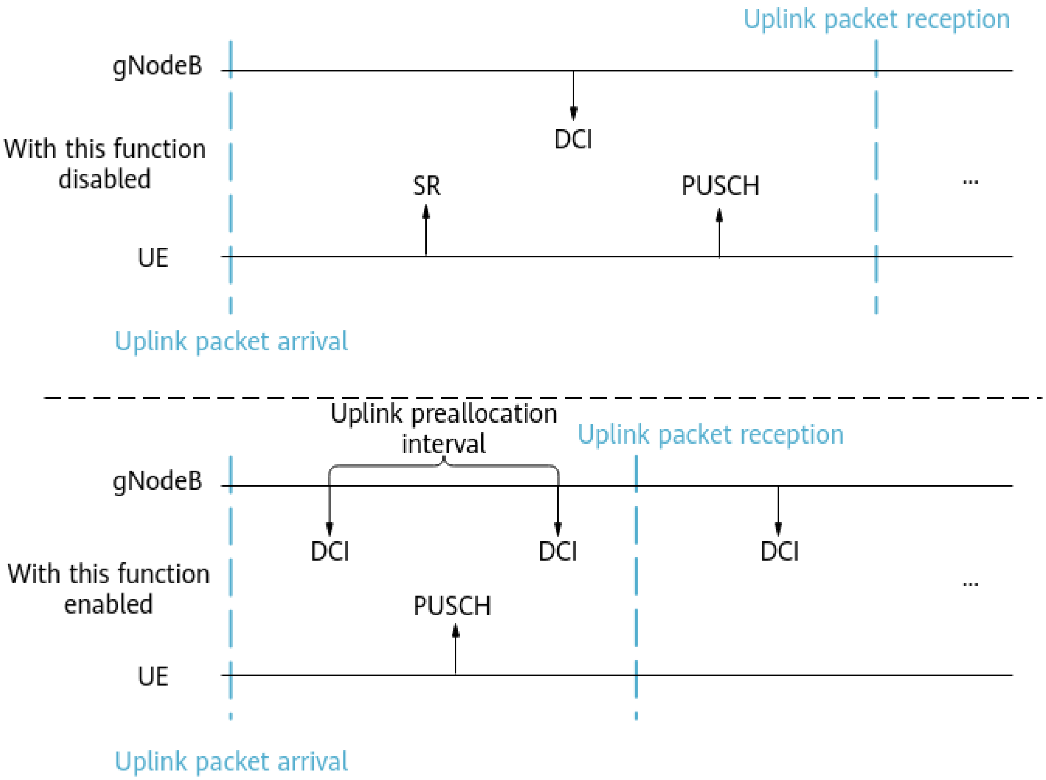
NOTE

For details about the cooperation between traditional uplink preallocation and QCI-based uplink preallocation, see **4.2.2 Impacts**.

QCI-based Uplink Basic Preallocation

In QCI-based uplink basic preallocation (see **Figure 4-1**), the scheduling of a UE persists as long as resources are available, regardless of whether the UE has service requests.

Figure 4-1 QCI-based uplink basic preallocation



This function is controlled by the **UL_PREALLOCATION_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter. After this function is enabled, the following parameters can be configured as required:

- **gNBDUSchParamGroup**.***UlPreallocationInterval***: interval of uplink preallocation
- **gNBDUSchParamGroup**.***UlPreallocationDataVolume***: data volume of uplink preallocation

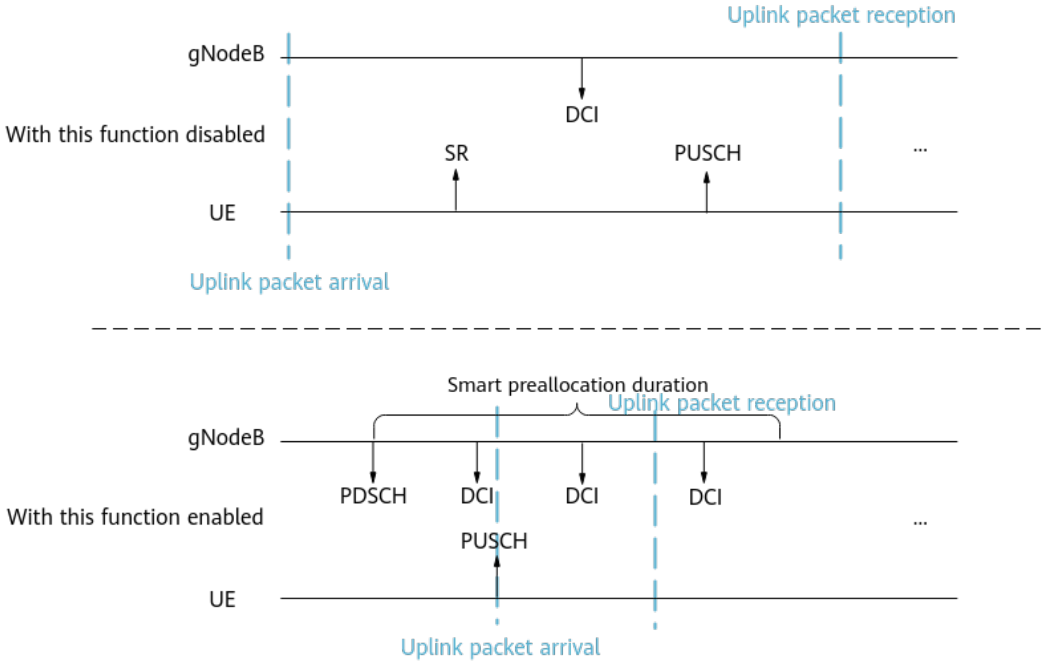
NOTE

If multiple QCI-specific bearers are set up for a UE and preallocation configurations differ among QCIs, the preallocation configuration associated with the QCI with the highest priority takes effect. For details about the QCI configuration and priority, see *QoS Management*.

QCI-based Uplink Smart Preallocation

In QCI-based uplink smart preallocation (see **Figure 4-2**), **the gNodeB continuously schedules UEs for a period of time only when the gNodeB has downlink data to transmit**. Therefore, QCI-based uplink smart preallocation saves on more uplink resources than QCI-based uplink basic preallocation whereby the gNodeB performs scheduling regularly.

Figure 4-2 QCI-based uplink smart preallocation



This function is controlled by the **UL_SMART_PREALLOC_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter. After this function is enabled, the following parameters can be configured as required:

- **gNBDUSchParamGroup**.***UlSmartPreallocationDur***: uplink smart preallocation duration
- **gNBDUSchParamGroup**.***UlPreallocationInterval***: interval of uplink preallocation
- **gNBDUSchParamGroup**.***UlPreallocationDataVolume***: data volume of uplink preallocation

If not all data is transmitted during the period specified by **gNBDUSchParamGroup**.***UlSmartPreallocationDur***, the duration for QCI-based uplink smart preallocation is automatically extended to allow all remaining data to be transmitted.

In addition, it is good practice to enable uplink preallocation UE priority retention to optimize the policy of retaining uplink smart preallocation priorities. This function is controlled by the **UL_PREALLOC_PRI_RETAIN_SW** option of the **NRDUCellUlSch**.***UlPreallocPolEnhSwitch*** parameter and takes effect only when QCI-based uplink smart preallocation is enabled.

NOTE

If multiple QCI-specific bearers are set up for a UE and preallocation configurations differ among QCIs, the preallocation configuration associated with the QCI with the highest priority takes effect. For details about the QCI configuration and priority, see *QoS Management*.

Implementation Policies of QCI-based Uplink Preallocation

For a UE with multiple QCI-specific bearers, the implementation of QCI-based uplink basic preallocation or QCI-based uplink smart preallocation depends on different policies described below:

- Policy 1: When the **SERV_DIFF_POLICY_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter is deselected, the switch control of QCI-based uplink basic preallocation and QCI-based uplink smart preallocation applies to the UE's bearers with the QCI of the highest priority. The functions do not take effect even if the function switches are turned on for the UE's bearers with other QCIs.
- (New policy for service differentiation) Policy 2: When the **SERV_DIFF_POLICY_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter is selected, QCI-based uplink basic preallocation or QCI-based uplink smart preallocation can take effect on the UE in accordance with specific principles as long as the function switch is turned on for any QCI-specific bearer of the UE. The aforementioned principles are as follows:
  - Interval of uplink preallocation: Among the preallocation intervals configured for all the QCIs for which QCI-based uplink basic preallocation or QCI-based uplink smart preallocation is enabled, the minimum interval of uplink preallocation applies.
  - Uplink preallocation data volume: Among the uplink preallocation data volumes configured for all the QCIs for which QCI-based uplink basic preallocation or QCI-based uplink smart preallocation is enabled, the maximum uplink preallocation data volume applies.
  - Uplink smart preallocation duration: Among the smart preallocation durations configured for all the QCIs for which QCI-based uplink smart preallocation is enabled, the maximum smart preallocation duration applies.

Policies for Controlling the Number of UEs for Preallocation

The policies for controlling the number of UEs for preallocation are introduced to allow the maximum numbers of UEs (1) for which QCI-based uplink preallocation can take effect and (2) for which cell-level uplink preallocation can take effect to be controlled by different parameters. This function is controlled by the **PRESCH_UE_NUM_CTRL_POL_SW** option of the **NRDUCellUlSch**.***UlPreallocPolEnhSwitch*** parameter.

- Policy 1: If this option is selected, the maximum numbers of UEs (1) for which QCI-based uplink preallocation can take effect and (2) for which cell-level uplink preallocation can take effect are separately configured.
  - The maximum number of UEs for which QCI-based uplink preallocation can take effect is specified by the **NRDUCellUlsch**.***QciPreschUeNumLimit*** parameter.
  - The maximum number of UEs for which cell-level uplink preallocation can take effect is specified by the **NRDUCellUlsch**.***CellPreschUeNumLimit*** parameter.
- Policy 2: If this option is deselected, the maximum number of UEs for which uplink preallocation can take effect is specified by the

NRDUCellPusch.***PreschUeNumUpperLimit*** parameter, irrespective of UE types.

NOTE

When the **PRESCH_UE_NUM_CTRL_POL_SW** option of the **NRDUCellUlSch**.***UlPreallocPolEnhSwitch*** parameter is selected, the maximum number of UEs for preallocation is not controlled by the **NRDUCellPusch**.***PreschUeNumUpperLimit*** parameter. However, it is recommended that the value of the **NRDUCellPusch**.***PreschUeNumUpperLimit*** parameter remain unchanged, thereby avoiding affecting other features and functions.

Frequency Selection Policy for QCI-based Preallocation

The **QCI_PRESCH_FREQ_SEL_POL_SW** option of the **NRDUCellUlSch**.***UlPreallocPolEnhSwitch*** specifies whether uplink frequency selective scheduling takes effect for UEs during QCI-based uplink preallocation. For details about uplink frequency selective scheduling, see *Uplink Scheduling* .

- Policy 1: If this option is selected, the base station checks whether the estimated RBs for UEs in QCI-based uplink preallocation are small packets. If yes, uplink frequency selective scheduling does not take effect for these UEs. This reduces the probability for small-packet UEs to interrupt the scheduling of large-packet UEs, thereby increasing the average uplink cell throughput.
- Policy 2: If this option is deselected, uplink frequency selective scheduling can take effect for UEs in QCI-based uplink preallocation. Specifically, these UEs can select the RBs with the optimal performance for scheduling, thereby reducing the uplink BER of these UEs.

4.1.2 QCI-specific SR Period Configuration

An SR is used to request resources in uplink scheduling. The SR period is configured on the network side for a UE through RRC signaling. For details, see *Channel Management*. **The QCI-specific SR period configuration function enables an SR period to be customized for a service carried on a QCI-specific bearer**. If multiple QCI-specific bearers are set up for a UE and SR period configuration differs between QCIs, the SR period configuration associated with the QCI with the highest priority takes effect.

This function is controlled by the **SR_PERIOD_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter. The SR period can be specified for a QCI using the **gNBDUSchParamGroup**.***SrPeriod*** parameter.

4.1.3 QCI-specific First-Packet Size Configuration for SR-based Scheduling

The "QCI-specific first-packet size configuration for SR-based scheduling" function enables the first-packet size to differ among QCI-specific bearers in SR-based scheduling, thereby reducing the delay in uplink packet transmissions. The **gNBDUSchParamGroup**.***SrSchFirstTbSize*** parameter can be configured to specify the first-packet size. The following table lists the settings of this parameter.

gNBDUSchParamGroup.*SrSchFirstTbSize*	**First-Packet Size**	**Setting Notes**
0 (default value)	The first-packet size for a QCI-specific bearer in SR-based scheduling takes the value of the **NRDUCellPusch**.***SrSchFirstTbSize*** parameter, which is a cell-level parameter.	None
A value ranging from 1 to 65535	The first-packet size for a QCI-specific bearer in SR-based scheduling takes the value of this parameter.	You are advised to set this parameter based on the actual service packet size. For example, if the service packet size is 500 bytes, you are advised to set this parameter to **4160** (in the unit of bit), considering 20 bytes of header overhead.

For a VoNR bearer, the first-packet size in SR-based scheduling complies with the VoNR baseline algorithm instead of being determined by the **gNBDUSchParamGroup**.***SrSchFirstTbSize*** parameter.

If multiple QCI-specific bearers are set up for a UE and the configurations of the first-packet size for SR-based scheduling differ among these bearers, the first-packet size configuration associated with the QCI with the highest priority takes effect.

4.1.4 QCI-specific Scheduling Based on the Target IBLER

In QCI-specific scheduling based on the target IBLER (see **Figure 4-3** and **Figure 4-4**), uplink and downlink target IBLERs can be configured for a given QCI. This configuration allows the IBLERs of services carried on bearers with that QCI to converge on the target IBLERs. The target IBLERs can be reduced to decrease the bit error rate (BER) and number of retransmissions, thereby decreasing the delay over the air interface.

QCI-specific scheduling based on the target IBLER includes QCI-specific scheduling based on the uplink target IBLER and QCI-specific scheduling based on the downlink target IBLER.

- QCI-specific scheduling based on the uplink target IBLER: After this function is enabled, the uplink target IBLER corresponding to the service-specific QCI is used during uplink scheduling. The **gNBDUSchParamGroup**.***UlTargetIbler*** parameter can be configured to specify an uplink target IBLER. This function is controlled by the **UL_IBLER_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter.
- QCI-specific scheduling based on the downlink target IBLER: After this function is enabled, the downlink target IBLER corresponding to the service-

specific QCI is used during downlink scheduling. The **gNBDUSchParamGroup**.***DlTargetIbler*** parameter can be configured to specify a downlink target IBLER. This function is controlled by the **DL_IBLER_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter.

If multiple QCI-specific bearers are set up for a UE and IBLER configurations differ among QCIs, the target IBLER configuration associated with the QCI with the highest priority takes effect.

When "QCI-specific service optimization for specific UEs" is enabled (by selecting the **SPEC_USR_SERVICE_OPT_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter):

- "QCI-specific scheduling based on the uplink target IBLER" takes effect only for UEs with a QCI-specific bearer which are not uplink full-buffer UEs.
- "QCI-specific scheduling based on the downlink target IBLER" takes effect only for UEs with a QCI-specific bearer which are not downlink full-buffer UEs.

Figure 4-3 QCI-specific scheduling based on the uplink target IBLER

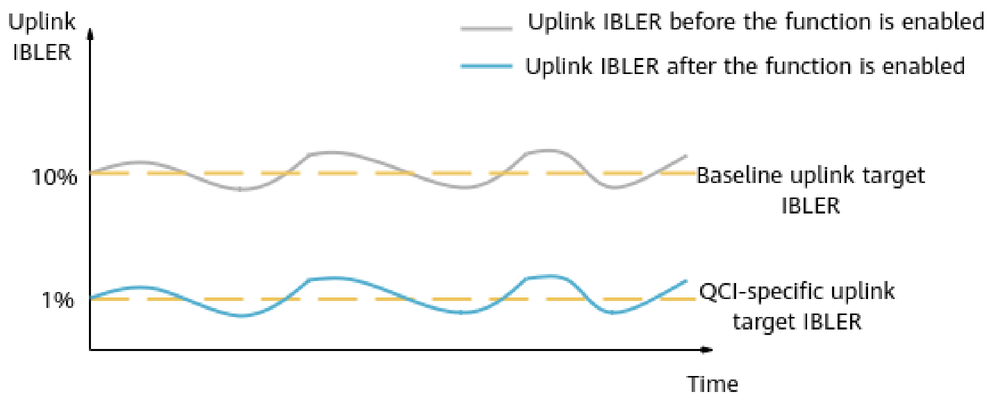
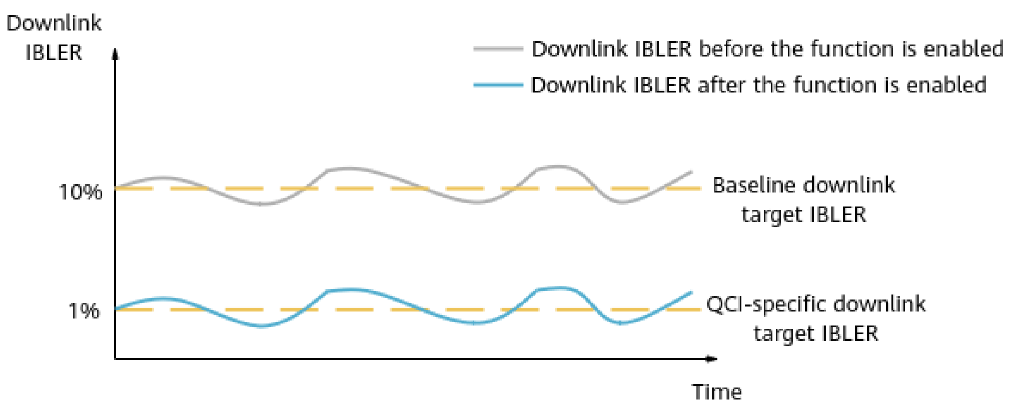


Figure 4-4 QCI-specific scheduling based on the downlink target IBLER



NOTE

If a QCI is not defined in the **NRDUCellQciBearer** MO, the downlink target IBLER for the QCI is determined by the **NRDUCellPdsch**.***DlTargetIbler*** parameter rather than the **gNBDUSchParamGroup**.***DlTargetIbler*** parameter.

4.2 Network Analysis

4.2.1 Benefits

Deployment Suggestions

It is recommended that a 5QI or QCI be planned separately for a service that requires differentiated guarantee and that the 5QI or QCI not be used for other services. On this basis, service-differentiated low latency functions are enabled for the QCI.

Benefits

Service-differentiated low latency functions provide the following benefits:

- After QCI-based uplink preallocation is enabled, the processing time required for scheduling is shortened compared with that in SR-based scheduling. The function benefits depend on the settings of the following parameters:
  - **gNBDUSchParamGroup**.***UlPreallocationDataVolume*** (indicating the data volume for uplink preallocation)
    - A larger value of this parameter results in a shorter first-packet delay of uplink data transmission, but stronger uplink interference and higher UE power consumption.
    - A smaller value of this parameter results in a longer first-packet delay of uplink data transmission, but weaker uplink interference and lower UE power consumption.
  - **gNBDUSchParamGroup**.***UlPreallocationInterval*** (indicating the uplink preallocation interval)
    - A smaller value of this parameter indicates a shorter uplink preallocation interval for UEs, which results in a shorter delay, but stronger uplink interference and higher UE power consumption.
    - A larger value of this parameter indicates a longer uplink preallocation interval for UEs, which results in weaker uplink interference and lower UE power consumption, but a longer delay.
- After QCI-specific SR period configuration is enabled, its benefits depend on the settings of the **gNBDUSchParamGroup**.***SrPeriod*** parameter.
  - A smaller value of this parameter indicates a shorter SR period, which results in a shorter E2E delay but also a smaller maximum number of online UEs supported in the cell.
  - A larger value of this parameter indicates a longer SR period, which results in a longer E2E delay but also a larger maximum number of online UEs supported in the cell.

- After QCI-specific first-packet size configuration for SR-based scheduling is enabled, its benefits depend on the settings of the **gNBDUSchParamGroup**.***SrSchFirstTbSize*** parameter.
  - A larger value of this parameter results in a shorter delay in uplink data transmissions but also higher UE power consumption.
  - A smaller value of this parameter results in lower UE power consumption but also a longer delay in uplink data transmissions.
- After QCI-specific scheduling based on the uplink target IBLER is enabled, a smaller value of the **gNBDUSchParamGroup**.***UlTargetIbler*** parameter results in fewer uplink transmissions over the air interface. This decreases the delay over the air interface when coverage is favorable and resources are sufficient. Whereas, a larger value of this parameter results in less resource consumption by uplink data transmissions and lower signal quality requirements.
- After QCI-specific scheduling based on the downlink target IBLER is enabled, a smaller value of the **gNBDUSchParamGroup**.***DlTargetIbler*** parameter results in fewer downlink transmissions over the air interface. This decreases the delay over the air interface when coverage is favorable and resources are sufficient. Whereas, a larger value of this parameter results in less resource consumption by downlink data transmissions and lower signal quality requirements.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

- QCI-based uplink preallocation
  - After this function is enabled, a larger maximum number of UEs for preallocation indicates a larger number of such UEs, while a smaller maximum number indicates a smaller number of such UEs. If a large number of UEs are involved in QCI-based preallocation, but a too small value is set for this parameter, the average uplink UE throughput will decrease and the E2E delay will increase. For details about the maximum number of UEs for preallocation, see **Policies for Controlling the Number of UEs for Preallocation** in **4.1.1 QCI-based Uplink Preallocation**.
  - When the new policy for service differentiation is used, the implementation of QCI-based uplink preallocation is related to configurations of all QCI-specific bearers of a UE. As a result, QCI-based uplink preallocation is more likely to take effect, while cell-level uplink preallocation is less likely to take effect. An increase in the overall probability that uplink preallocation takes effect (dependent on the uplink preallocation interval and the number of UEs for which uplink preallocation takes effect) will result in an increase in the uplink PRB usage and average uplink cell throughput. However, this will result in an increase in interference to neighboring cells.
  - When the switch for "frequency selection policy for QCI-based preallocation" is turned on, the uplink cell throughput and uplink user-perceived rate can increase. However, this function may increase the uplink BLER of UEs with QCI-based uplink preallocation in effect, thereby increasing the E2E delay.

- QCI-specific SR period configuration

  The impacts of this function depend on the setting of the **gNBDUSchParamGroup**.***SrPeriod*** parameter.

  - A smaller value of this parameter indicates a shorter SR period, which results in a shorter E2E delay but also a smaller maximum number of online UEs supported in the cell.
  - A larger value of this parameter indicates a longer SR period, which results in a longer E2E delay but also a larger maximum number of online UEs supported in the cell.

- QCI-specific first-packet size configuration for SR-based scheduling

  The impacts of this function depend on the setting of the **gNBDUSchParamGroup**.***SrSchFirstTbSize*** parameter.

  - A larger value of this parameter results in a shorter delay in uplink data transmissions but also higher UE power consumption.
  - A smaller value of this parameter results in lower UE power consumption but also a longer delay in uplink data transmissions.

- QCI-specific scheduling based on the target IBLER
  - After QCI-specific scheduling based on the uplink target IBLER is enabled, its impacts depend on the settings of the **gNBDUSchParamGroup**.***UlTargetIbler*** parameter. A smaller value of this parameter results in higher resource consumption by uplink data transmissions and higher signal quality requirements. Whereas, a larger value of this parameter results in a larger number of uplink transmissions over the air interface and a longer delay over the air interface.
  - After QCI-specific scheduling based on the downlink target IBLER is enabled, its impacts depend on the settings of the **gNBDUSchParamGroup**.***DlTargetIbler*** parameter. A smaller value of this parameter results in higher resource consumption by downlink data transmissions and higher signal quality requirements. Whereas, a larger value of this parameter results in a larger number of downlink transmissions over the air interface and a longer delay over the air interface.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-050208	Service-Differentiated Low Latency	NR0S00PSCH00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SR period adaptation	**SR_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch**.***SrResoureAlgoSwitch*** parameter	*Channel Management*	When **NRDUCell**.***DuplexMode*** is set to **CELL_SUL** or **CELL_SDL**, SR period adaptation and QCI-specific SR period configuration are mutually exclusive.
Low-frequency TDD	IAB deployment	**NRDUCELL**.***DeployPosition*** set to **DEPLOY_IAB**	None	IAB deployment and service-differentiated low latency are mutually exclusive.
Low-frequency TDD	PLC UE number specification improvement	**UL_PREALLOC_SELFSTUDY_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter	*URLLC*	QCI-based uplink smart preallocation and PLC UE number specification improvement are mutually exclusive.
Low-frequency TDD	QCI-based uplink basic preallocation	**UL_PREALLOCATION_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter	*Service-differentiated Experience Guarantee*	QCI-based uplink basic preallocation and QCI-based uplink smart preallocation cannot be enabled at the same time.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-based uplink smart preallocation	**UL_SMART_PREALLOC_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter	*Service-differentiated Experience Guarantee*	

4.3.2.3 Function Impacts

- QCI-based uplink preallocation

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink preallocation for sparse-packet services	**SPARSE_DIFF_PROC_SW** option of the **gNodeBParam.*ServiceDiffProcSwitch*** parameter and **SPARSE_DIFF_SCH_SW** option of the **NRDUCellAlgoSwitch.*ServiceDiffSwitch*** parameter	*Uplink Scheduling*	QCI-based uplink basic preallocation and QCI-based uplink smart preallocation do not take effect for UEs enabled with uplink preallocation for sparse-packet services.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink basic preallocation	**UL_BASIC_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter	*Uplink Scheduling*	For a service with a specific QCI, when both uplink basic preallocation and QCI-based uplink basic preallocation are enabled, the impacts are as follows: ● If the **SERV_DIFF_POLICY_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter is selected, uplink basic preallocation does not take effect. ● If the **SERV_DIFF_POLICY_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter is deselected, both functions can take effect at the same time but are mutually exclusive at the level of TTI. To illustrate, within a TTI, QCI-based uplink basic preallocation preferentially takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink smart preallocation	**UL_SMART_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter	*Uplink Scheduling*	For a service with a specific QCI, when both uplink smart preallocation and QCI-based uplink smart preallocation are enabled, the impacts are as follows: ● If the **SERV_DIFF_POLICY_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter is selected, uplink smart preallocation does not take effect. ● If the **SERV_DIFF_POLICY_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter is deselected, both functions can take effect at the same time but are mutually exclusive at the level of TTI. To illustrate, within a TTI, QCI-based uplink smart preallocation preferentially takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	For CA UEs, QCI-based uplink basic preallocation and QCI-based uplink smart preallocation take effect only in their primary component carriers (PCCs), not secondary component carriers (SCCs).
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	For CA UEs, QCI-based uplink basic preallocation and QCI-based uplink smart preallocation take effect only in their PCCs, not SCCs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DRX	**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*DRX*	You are advised not to enable DRX for bearers with QCI-based uplink preallocation enabled. When DRX is enabled together with QCI-based uplink basic preallocation or QCI-based uplink smart preallocation, the impacts are as follows: ● If there is no voice service, the actual uplink preallocation interval (specified by the **gNBDUSchParamGroup**.***UlPreallocationInterval*** parameter) may be less than the configured value. ● If there are voice services, the actual uplink preallocation interval (specified by the **gNBDUSchParamGroup**.***UlPreallocationInterval*** parameter) is the same as the configured value.
Low-frequency TDD	Device-pipe synergy preallocation	**UL_PREFERENTIAL_SCH_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter	*FWA*	QCI-based uplink preallocation does not take effect for UEs with device-pipe synergy preallocation in effect.

- QCI-specific SR period configuration

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SR period adaptation	**SR_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch**.***SrResoureAlgoSwitch*** parameter	*Channel Management*	When both SR period adaptation and QCI-specific SR period configuration are enabled, the **gNBDUSchParamGroup**.***SrPeriod*** parameter can only be set to **20SLOT**, **40SLOT**, or **80SLOT**.

- QCI-specific scheduling based on the target IBLER

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink preallocation for sparse-packet services	**SPARSE_DIFF_PROC_SW** option of the **gNodeBParam.*ServiceDiffProcSwitch*** parameter and **SPARSE_DIFF_SCH_SW** option of the **NRDUCellAlgoSwitch.*ServiceDiffSwitch*** parameter	*Uplink Scheduling*	QCI-specific scheduling based on the uplink target IBLER and QCI-specific scheduling based on the downlink target IBLER do not take effect for UEs enabled with uplink preallocation for sparse-packet services.
Low-frequency TDD	DRX	**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*DRX*	You are advised not to enable DRX for bearers with the following functions enabled: QCI-specific scheduling based on the uplink target IBLER and QCI-specific scheduling based on the downlink target IBLER.

- QCI-specific first-packet size configuration for SR-based scheduling

  None

4.3.3 Hardware

The hardware described in this section may require compatibility between each other. For details, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards, as well as the NR-only UBBPg series and later baseband processing units, support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

In NR TDD, all NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Cells

- When the SCS of a cell is 15 kHz, the QCI-specific SR period (**gNBDUSchParamGroup**.***SrPeriod***) configured in the cell cannot be greater than **80SLOT**.
- Service-differentiated low latency cannot be enabled if all the following conditions are met: (1) **NRDUCell**.***FrequencyBand*** is set to **N257**, **N258**, **N260**, or **N261**; (2) **NRDUCell**.***DuplexMode*** is set to **CELL_TDD**; (3) **NRDUCell**.***SlotAssignment*** is set to **4_1_DDDSU**; (4) **NRDUCell**.***SubcarrierSpacing*** is set to **120KHZ**; (5) **NRDUCell**.***CellRadius*** is set to a value greater than **2410**.

4.3.5 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Low Latency Switch	**NRDUCellAlgoSwitch**.***LowLatencySwitch***	SERV_DIFF_LOW_LATENCY_SW	To enable service-differentiated low latency, select this option.
Low Latency Switch	**NRDUCellAlgoSwitch**.***LowLatencySwitch***	SERV_DIFF_POLICY_SW	To enable the new policy for service differentiation, select this option.
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	UL_PREALLOCATION_QCI_SW	To enable QCI-based uplink preallocation, select this option. QCI-based uplink preallocation and QCI-based uplink smart preallocation cannot be enabled simultaneously.
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	UL_IBLER_QCI_SW	To enable QCI-specific scheduling based on the uplink target IBLER, select this option.
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	DL_IBLER_QCI_SW	To enable QCI-specific scheduling based on the downlink target IBLER, select this option.
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	UL_SMART_PREALLOC_QCI_SW	To enable QCI-based uplink smart preallocation, select this option. QCI-based uplink smart preallocation and QCI-based uplink preallocation cannot be enabled simultaneously.
UL Preallocation Policy Enhancement Switch	**NRDUCellUlSch**.***UlPreallocPolEnhSwitch***	UL_PREALLOC_PRI_RETAIN_SW	You are advised to select this option when QCI-based uplink smart preallocation is enabled.

Parameter Name	Parameter ID	Option	Setting Notes
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	SR_PERIOD_QCI_SW	To enable QCI-specific SR period configuration, select this option.
Uplink Preallocation Interval	**gNBDUSchParamGroup**.***UlPreallocationInterval***	None	After QCI-based uplink preallocation is enabled, set this parameter based on the network plan.
Uplink Preallocation Data Volume	**gNBDUSchParamGroup**.***UlPreallocationDataVolume***	None	After QCI-based uplink preallocation is enabled, set this parameter based on the network plan.
Uplink Smart Preallocation Duration	**gNBDUSchParamGroup**.***UlSmartPreallocationDur***	None	After QCI-based uplink smart preallocation is enabled, set this parameter based on the network plan.
UL Preallocation Policy Enhancement Switch	**NRDUCellUlSch**.***UlPreallocPolEnhSwitch***	PRESCH_UE_NUM_CTRL_POL_SW	Select this option if the maximum numbers of UEs (1) for which QCI-based uplink preallocation can take effect and (2) for which cell-level uplink preallocation can take effect need to be separately configured.
QCI Prescheduled UE Num Limit	**NRDUCellUlSch**.***QciPreschUeNumLimit***	None	Set this parameter based on the network plan when the **PRESCH_UE_NUM_CTRL_POL_SW** option of the **NRDUCellUlSch**.***UlPreallocPolEnhSwitch*** parameter is selected.

Parameter Name	Parameter ID	Option	Setting Notes
Cell Prescheduled UE Num Limit	**NRDUCellUlSch**.***CellPreschUeNumLimit***	None	Set this parameter based on the network plan when the **PRESCH_UE_NUM_CTRL_POL_SW** option of the **NRDUCellUlSch**.***UlPreallocPolEnhSwitch*** parameter is selected.
UL Preallocation Policy Enhancement Switch	**NRDUCellUlSch**.***UlPreallocPolEnhS-witch***	QCI_PRESCH_FREQ_SEL_POL_SW	Deselect this option. Select this option if (1) QCI-based uplink preallocation is enabled, (2) the cell's uplink load is high, and (3) the impact of UEs with QCI-based uplink preallocation in effect on the cell's uplink performance needs to be alleviated.
SR Period	**gNBDUSchParamGroup**.***SrPeriod***	None	After QCI-specific SR period configuration is enabled, you are advised to set this parameter based on the actual service packet size. For example, if the service packet size is 500 bytes, you are advised to set this parameter to **4160** (in the unit of bit), considering 20 bytes of header overhead.
SR-based Scheduling First TB Size	**gNBDUSchParamGroup**.***SrSchFirstTbSize***	None	After QCI-specific first-packet size configuration for SR-based scheduling is enabled, set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink Target IBLER	**gNBDUSchParamGroup**.***UlTargetIbler***	None	After QCI-specific scheduling based on the uplink target IBLER is enabled, set this parameter based on the network plan.
Downlink Target IBLER	**gNBDUSchParamGroup**.***DlTargetIbler***	None	After QCI-specific scheduling based on the downlink target IBLER is enabled, set this parameter based on the network plan.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

This function takes effect 5 to 10 minutes after the relevant parameter configurations in the **gNBDUSchParamGroup** and **NRDUCellQciBearer** MOs are modified.

Activation Command Examples

Basic configuration for service-differentiated low latency

```
//Turning on the switch for service-differentiated low latency
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, LowLatencySwitch=SERV_DIFF_LOW_LATENCY_SW-1;
//(Optional) Enabling the new policy for service differentiation
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, LowLatencySwitch=SERV_DIFF_POLICY_SW-1;
//Adding a gNodeB DU scheduling parameter group
ADD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1;
//Mapping the gNodeB DU scheduling parameter group to a bearer of a specific QCI
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, ScheduleParamGroupId=1;
```

QCI-based uplink preallocation (QCI-based uplink preallocation and QCI-based uplink smart preallocation cannot be enabled at the same time.)

- Enabling QCI-based uplink preallocation

```
//Turning on the switch for QCI-based uplink preallocation
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_PREALLOCATION_QCI_SW-1;
//Modifying the interval and data volume for uplink preallocation
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, UlPreallocationInterval=5,
UlPreallocationDataVolume=100;
```

- Enabling QCI-based uplink smart preallocation

```
//Turning on the switch for QCI-based uplink smart preallocation
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_SMART_PREALLOC_QCI_SW-1;
//Modifying the interval and data volume for uplink preallocation and the duration for uplink smart
preallocation
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, UlPreallocationInterval=5,
UlPreallocationDataVolume=100,UlSmartPreallocationDur=250;
//Enabling the uplink preallocation UE priority retention function
MOD NRDUCELLULSCH: NrDuCellId=1, UlPreallocPolEnhSwitch=UL_PREALLOC_PRI_RETAIN_SW-1;
```

- (Optional) Configuring a policy for controlling the number of UEs for preallocation

```
//Selecting the PRESCH_UE_NUM_CTRL_POL_SW option if the maximum numbers of UEs (1)  for
which QCI-based uplink preallocation can take effect and (2) for which cell-level uplink preallocation
can take effect need to be separately configured.
MOD NRDUCELLULSCH: NrDuCellId=1, UlPreallocPolEnhSwitch=PRESCH_UE_NUM_CTRL_POL_SW-1;
//Reconfiguring the QCI Prescheduled UE Num Limit and Cell Prescheduled UE Num Limit parameters
MOD NRDUCELLULSCH: NrDuCellId=1, QciPreschUeNumLimit=2, CellPreschUeNumLimit=2;
```

- (Optional) Configuring a frequency selection policy for QCI-based preallocation

```
//Turning on the switch for "frequency selection policy for QCI-based preallocation" when the cell's
uplink load is high and the impact of UEs with QCI-based uplink preallocation in effect on the cell's
uplink performance needs to be alleviated
MOD NRDUCELLULSCH: NrDuCellId=1, UlPreallocPolEnhSwitch=QCI_PRESCH_FREQ_SEL_POL_SW-1;
```

QCI-specific SR period configuration

```
//Turning on the switch for QCI-specific SR period configuration
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=SR_PERIOD_QCI_SW-1;
//Modifying the SR period
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, SrPeriod=20SLOT;
```

QCI-specific first-packet size configuration for SR-based scheduling

```
//Modifying the size of the first packet of a specific QCI in SR-based scheduling
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, SrSchFirstTbSize=600;
```

QCI-specific scheduling based on the uplink target IBLER

```
//Turning on the switch for QCI-specific scheduling based on the uplink target IBLER
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=UL_IBLER_QCI_SW-1;
//Changing the QCI-based uplink target IBLER to 10%
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, UlTargetIbler=10000;
```

QCI-specific scheduling based on the downlink target IBLER

```
//Turning on the switch for QCI-specific scheduling based on the downlink target IBLER
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=DL_IBLER_QCI_SW-1;
//Changing the QCI-based downlink target IBLER to 10%
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, DlTargetIbler=10000;
//Mapping the gNodeB DU scheduling parameter group to the QCI-specific bearer
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, ScheduleParamGroupId=1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling QCI-based uplink preallocation
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_PREALLOCATION_QCI_SW-0;

//Disabling QCI-based uplink smart preallocation and uplink preallocation UE priority retention
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_SMART_PREALLOC_QCI_SW-0;
MOD NRDUCELLULSCH: NrDuCellId=1, UlPreallocPolEnhSwitch=UL_PREALLOC_PRI_RETAIN_SW-0;

//Deselecting the QCI_PRESCH_FREQ_SEL_POL_SW option
MOD NRDUCELLULSCH: NrDuCellId=1, UlPreallocPolEnhSwitch=QCI_PRESCH_FREQ_SEL_POL_SW-0;

//Deselecting the PRESCH_UE_NUM_CTRL_POL_SW option
MOD NRDUCELLULSCH: NrDuCellId=1, UlPreallocPolEnhSwitch=PRESCH_UE_NUM_CTRL_POL_SW-0;

//(Optional) Disabling the new policy for service differentiation
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, LowLatencySwitch=SERV_DIFF_POLICY_SW-0;

//Disabling QCI-specific SR period configuration
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=SR_PERIOD_QCI_SW-0;

//Disabling QCI-specific first-packet size configuration for SR-based scheduling
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, SrSchFirstTbSize=0;

//Disabling QCI-specific scheduling based on the uplink target IBLER
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=UL_IBLER_QCI_SW-0;

//Disabling QCI-specific scheduling based on the downlink target IBLER
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=DL_IBLER_QCI_SW-0;

//Removing the mapping between the gNodeB DU scheduling parameter group and the bearer of the
specific QCI
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, ScheduleParamGroupId=255;

//Removing the gNodeB DU scheduling parameter group
RMV GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1;

//Turning off the switch for service-differentiated low latency
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, LowLatencySwitch=SERV_DIFF_LOW_LATENCY_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

On the MAE-Access, perform the following steps to check whether QCI-based uplink preallocation has taken effect:

Step 1 Enable a UE to access a cell. After a bearer with a specific QCI or 5QI is set up, move the UE to the cell center and enable the UE to perform a downlink ping service and stop performing uplink services.

Step 2 On the MAE-Access, start UE-level channel quality monitoring (corresponding to **Quality of Channel Monitoring** on the GUI) and MCS-based count monitoring (corresponding to **MCS Count Monitoring** on the GUI).

Step 3 Observe the UE channel quality by **Quality of Channel Monitoring** and the number of uplink scheduling times by **MCS Count Monitoring**. If uplink scheduling is implemented in each scheduling period, QCI-based uplink preallocation has taken effect.

----End

On the MAE-Access, perform the following steps to check whether the functions of QCI-specific scheduling based on the uplink and downlink target IBLERs have taken effect:

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 In the left navigation pane of the displayed **Signaling Trace Management** tab page, choose **User Performance Monitoring** > **BLER Monitoring** to observe the BLER.

----End

4.4.3 Network Monitoring

The indicators listed in **4.2 Network Analysis** can be used to monitor the benefits and impacts of this function.

5 Service-differentiated Link Reliability

5.1 Principles

Different QCI-specific bearers have different reliability requirements for services. Against this backdrop, service-differentiated link reliability is introduced to enhance the link reliability for UEs with QCI-specific bearers. **This function enables differentiated scheduling parameter configurations among UEs with different QCIs, reducing the BLER and number of retransmissions. This improves link reliability.** Service-differentiated link reliability is controlled by the **LINK_RELIABILITY_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter and includes the following sub-functions.

Sub-function Name	Description	Reference
QCI-specific PDCCH aggregation level optimization	For UEs with QCI-specific bearers, the PDCCH SINR and PDCCH BLER are adjusted, and the PDCCH aggregation level is adaptively adjusted based on the PUSCH DTX status, reducing the probability of DCI miss-detection. In this way, the uplink packet loss rate is reduced.	**5.1.1 QCI-specific PDCCH Aggregation Level Optimization**
QCI-specific uplink MCS selection optimization	The uplink MCS indexes for initial transmissions of UEs with QCI-specific bearers are reduced to decrease the uplink packet loss rate.	**5.1.2 QCI-specific Uplink MCS Selection Optimization**

Sub-function Name	Description	Reference
QCI-specific downlink MCS selection optimization	For UEs with QCI-specific bearers, the gNodeB decreases the CQI by a fixed value (1, 2, or 3) if HARQ feedback for initial transmission is NACK. This reduces the downlink MCS index, thereby lowering the downlink packet loss rate.	**5.1.3 QCI-specific Downlink MCS Selection Optimization**
QCI-specific uplink MU pairing differentiation	This function prohibits the UEs with QCI-specific bearers from participating in PUSCH MU-MIMO pairing, which reduces inter-layer interference of pairing.	**5.1.4 QCI-specific Uplink MU Pairing Differentiation**
QCI-specific downlink MU pairing assurance	This function lowers the correlation threshold for downlink MU-MIMO pairing between UEs with QCI-specific bearers and other UEs. This makes pairing between them more difficult and mitigates inter-layer interference of pairing.	**5.1.5 QCI-specific Downlink MU Pairing Assurance**
QCI-specific downlink power control optimization	The PDSCH PSD is adaptively adjusted based on the remaining power after scheduling, thereby achieving downlink power compensation for UEs with QCI-specific bearers.	**5.1.6 QCI-specific Downlink Power Control Optimization**
QCI-specific uplink power control optimization	If there is remaining uplink transmit power for UEs with QCI-specific bearers at the cell center or at a medium distance from the cell center, the PUSCH PSD can be increased to improve link reliability.	**5.1.7 QCI-specific Uplink Power Control Optimization**
Differentiated gap-assisted measurement	This function specifies whether UEs with QCI-specific bearers participate in gap-assisted measurement.	**5.1.8 Differentiated Gap-assisted Measurement**

Sub-function Name	Description	Reference
Explicit scheduling for downlink second retransmission	For UEs with QCI-specific bearers, if the base station detects that the first-retransmission result reported by the UEs is NACK and the TBS allowed at the second retransmission is greater than or equal to the TBS allowed at the initial transmission, the base station uses explicit scheduling for retransmission. That is, the base station sends the DCI used for retransmission to the UEs to indicate a specific MCS index. This increases the demodulation success rate of the UEs.	**5.1.9 Explicit Scheduling for Downlink Second Retransmission**

NOTE

All sub-functions require the **LINK_RELIABILITY_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter to be selected, except QCI-specific uplink power control optimization, QCI-specific uplink MU pairing differentiation, and differentiated gap-assisted measurement.

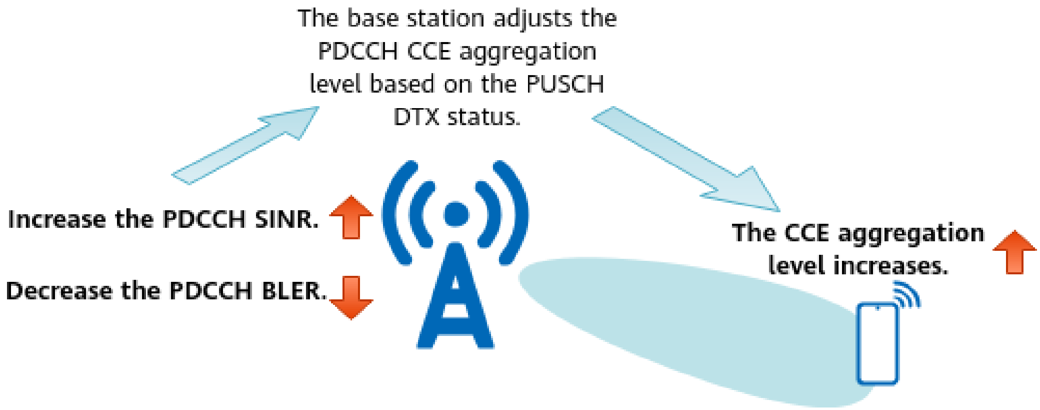
5.1.1 QCI-specific PDCCH Aggregation Level Optimization

A low PDCCH CCE aggregation level for a CEU increases the probability of DCI miss-detection for the corresponding service, causing downlink packet loss. As shown in **Figure 5-1**, QCI-specific PDCCH aggregation level optimization incorporates certain optimizations to increase the PDCCH CCE aggregation levels for UEs with QCI-specific bearers, thereby reducing the probability of DCI miss-detection. As a result, packet losses decrease and user experience improves. Such optimizations include:

- Optimization 1: **The PDCCH aggregation levels for UEs with QCI-specific bearers are adaptively adjusted based on the PUSCH DTX status.** This optimization is controlled by the **PUSCH_DTX_BASED_PDCCH_ADJ_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter.
- Optimization 2: **The PDCCH SINR for UEs with QCI-specific bearers can be adjusted.** On the gNodeB, an offset can be configured for the PDCCH SINR of the PDCCH CCE aggregation level for UEs with QCI-specific bearers, thereby reducing the PDCCH SINR of these UEs. The SINR offset is specified by the **gNBDUSchParamGroup**.***PdcchSinrOffset*** parameter. A larger value of this parameter results in a higher PDCCH CCE aggregation level.
- Optimization 3: **The target PDCCH BLER can be adjusted for UEs with QCI-specific bearers.** This optimization is controlled by the **DL_BLER_TAR_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter. After this option is

selected, a target PDCCH BLER can be configured on the gNodeB for UEs with QCI-specific bearers through the **gNBDUSchParamGroup**.***PdcchBlerTarget*** parameter. A smaller value of this parameter results in a higher PDCCH CCE aggregation level.

Figure 5-1 QCI-specific PDCCH aggregation level optimization



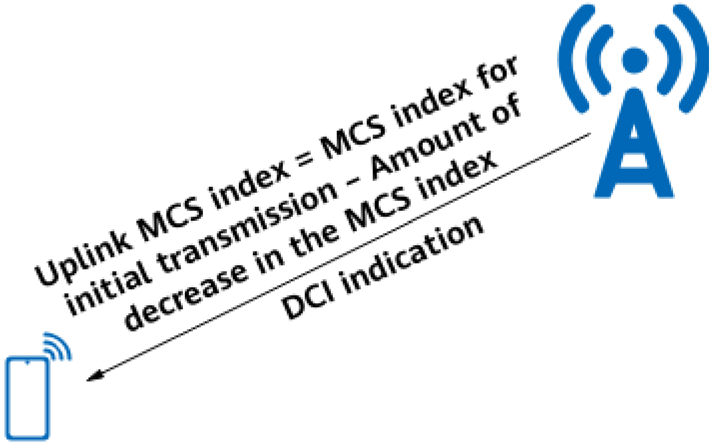
NOTE

- Optimization 1 does not apply to VoNR UEs (5QI 1 bearers) and ViNR UEs (5QI 2 bearers). For details about PDCCH aggregation level optimization for these two kinds of UEs, see section "PDCCH CCE Aggregation Level Increase" in *VoNR* and *ViNR*.
- The **gNBDUSchParamGroup**.***PdcchSinrOffset*** parameter in optimization 2 does not apply to VoNR UEs (5QI 1 bearers). The PDCCH SINR offset for VoNR UEs takes effect based on the **NRDUCellPdcchAlgo**.***VoicePdcchSinrOffset*** parameter setting. For details, see section "PDCCH CCE Aggregation Level Increase" in *VoNR*.
- When multiple bearers are set up for a UE, the function parameter settings corresponding to the UE's QCI with the highest priority take effect. For details about QCI configurations of UEs and QCI priorities, see *QoS Management*.

5.1.2 QCI-specific Uplink MCS Selection Optimization

A high uplink MCS index increases the uplink packet loss rate. **As shown in Figure 5-2, QCI-specific uplink MCS selection optimization decreases the uplink MCS indexes for initial transmissions of UEs with QCI-specific bearers**, thereby reducing the uplink packet loss rate and improving user experience.

Figure 5-2 QCI-specific uplink MCS selection optimization



This function is controlled by the **SCH_MCS_SELECT_OPT_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter, and the **gNBDUSchParamGroup**.***UlInitTransMcsDecrease*** parameter specifies the decrease in the MCS index for uplink initial transmission of UEs.

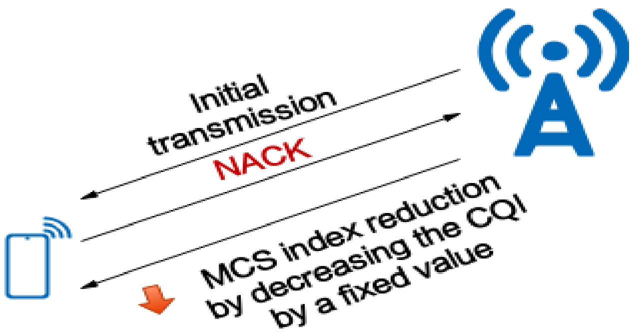
NOTE

When multiple bearers are set up for a UE, the function parameter settings corresponding to the UE's QCI with the highest priority take effect. For details about QCI configurations of UEs and QCI priorities, see *QoS Management*.

5.1.3 QCI-specific Downlink MCS Selection Optimization

A high downlink MCS index increases the downlink packet loss rate. **As shown in Figure 5-3, after QCI-specific downlink MCS selection optimization is enabled, the gNodeB decreases the CQI value by a fixed value (1, 2, or 3) to reduce the downlink MCS index when the HARQ feedback is NACK for the initial transmissions of UEs with QCI-specific bearers. The MCS index is gradually reduced to a value that matches the channel quality by decreasing the CQI value several times,** thereby improving transmission reliability.

Figure 5-3 QCI-specific downlink MCS selection optimization



This function is controlled by the **SCH_MCS_SELECT_OPT_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter, and the

gNBDUSchParamGroup.***DlCqiAdjValue*** parameter specifies the downlink CQI adjustment value.

NOTE

- This function does not apply to VoNR UEs (5QI 1 bearers) or ViNR UEs (5QI 2 bearers). For details about the related function for VoNR UEs (5QI 1 bearers), see section "Fast MCS Index Reduction in the Downlink" in *VoNR*. For details about the related function for ViNR UEs (5QI 2 bearers), see section "Optimized Downlink MCS Index Selection for Initial Transmission" in *ViNR*.
- When multiple bearers are set up for a UE, the function parameter settings corresponding to the UE's QCI with the highest priority take effect. For details about QCI configurations of UEs and QCI priorities, see *QoS Management*.

5.1.4 QCI-specific Uplink MU Pairing Differentiation

Uplink UE pairing increases the inter-UE interference and BER and reduces the link reliability of UEs. To address this issue, QCI-specific uplink MU pairing differentiation is introduced. This function enables differentiated configurations of whether UEs with QCI-specific bearers participate in PUSCH MU-MIMO pairing.

- When the **UL_MU_MIMO_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter is deselected for QCI-specific bearers, **UEs with QCI-specific bearers are not allowed to participate in PUSCH MU-MIMO pairing, reducing inter-layer interference of pairing and improving the link reliability of these UEs.**
- When the **UL_MU_MIMO_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter is selected for QCI-specific bearers, UEs with QCI-specific bearers are allowed to participate in PUSCH MU-MIMO pairing.

NOTE

- Assume that multiple bearers are set up for a UE. If the **UL_MU_MIMO_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter is deselected for any QCI of the UE, the UE does not participate in PUSCH MU-MIMO pairing.
- For QCI 1 bearers, the **UL_MU_MIMO_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter must be deselected.

5.1.5 QCI-specific Downlink MU Pairing Assurance

UE pairing increases the inter-UE interference and BER and reduces the link reliability of UEs. To address this issue, QCI-specific downlink MU pairing assurance is introduced. **This function lowers the correlation threshold for downlink MU-MIMO pairing between UEs with QCI-specific bearers and other UEs. This makes pairing between them more difficult and mitigates inter-layer interference of pairing, thereby improving the link reliability of specific UEs.**

This function is controlled by the **DL_MU_MIMO_CORR_THLD_ADJ_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter.

NOTE

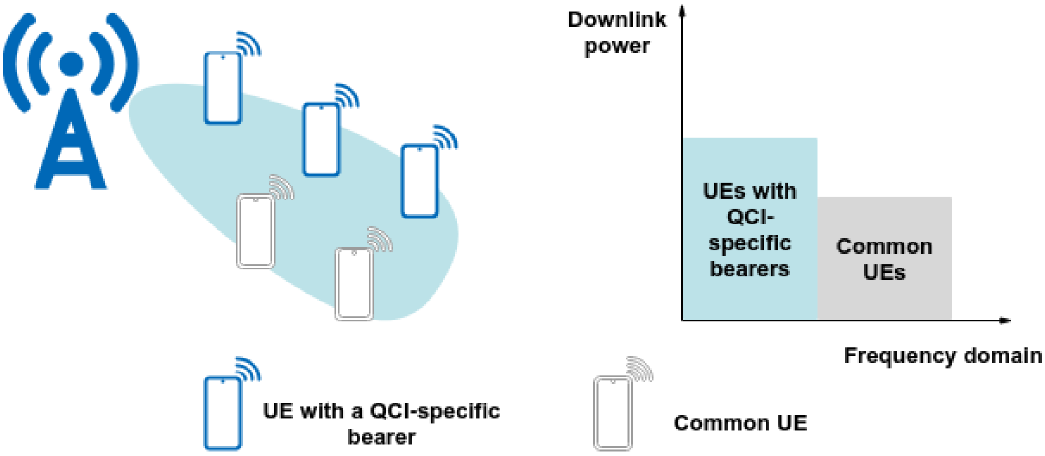
When multiple bearers are set up for a UE, the function parameter settings corresponding to the UE's QCI with the highest priority take effect. For details about QCI configurations of UEs and QCI priorities, see *QoS Management*.

5.1.6 QCI-specific Downlink Power Control Optimization

After scheduling, the gNodeB can aggregate the remaining power for specific UEs to reduce bit errors and retransmissions, thereby improving link reliability. QCI-specific downlink power control optimization is introduced to support downlink power compensation for UEs with QCI-specific bearers.

As shown in Figure 5-4, this function enables the gNodeB to adaptively adjust the PDSCH PSD based on the remaining power after scheduling, thereby achieving downlink power compensation for UEs with QCI-specific bearers. In this way, user experience of these UEs can be improved. In addition, the gNodeB keeps the increase in the PDSCH power per RB within 3 dB to avoid an excessively high PSD.

Figure 5-4 QCI-specific downlink power control optimization



This function is controlled by the **DL_PWR_AGG_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter.

NOTE

QCI-specific downlink power control optimization cannot be enabled for bearers with QCI 1, 2, 65, or 66.

When multiple bearers are set up for a UE, the function parameter settings corresponding to the UE's QCI with the highest priority take effect. For details about QCI configurations of UEs and QCI priorities, see *QoS Management*.

5.1.7 QCI-specific Uplink Power Control Optimization

If there is remaining uplink transmit power for UEs with QCI-specific bearers at the cell center or at a medium distance from the cell center, the PUSCH PSD can be increased to improve link reliability. QCI-specific uplink power control optimization is introduced to adjust uplink power control parameter configurations for UEs with QCI-specific bearers.

This function is controlled by the **UL_PWR_CTRL_CONFIG_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter. **After this function is enabled, the following power control parameters can be configured for UEs with QCI-specific bearers and the QCI-level parameter settings take effect in**

uplink power control for the UEs with the associated QCI. Otherwise, cell-level parameter settings take effect.

Parameter Name	QCI-Level Parameter	Cell-Level Parameter
PUSCH Outer-Loop Power Control Target	**gNBDuSchParamGroup**.***PuschOuterLoopPcTarget***	**NRDUCellUlPcConfig**.***PuschOuterLoopPcTarget***
Near Point PUSCH SINR Threshold Offset	**gNBDuSchParamGroup**.***NearPointPuschSinrThldOfs***	**NRDUCellUlPcConfig**.***NearPointPuschSinrThldOfs***
PUSCH RSRP Threshold	**gNBDuSchParamGroup**.***PuschRsrpThld***	**NRDUCellUlPcConfig**.***PuschRsrpThld***

NOTE

When multiple bearers are set up for a UE, the function parameter settings corresponding to the UE's QCI with the highest priority take effect. For details about QCI configurations of UEs and QCI priorities, see *QoS Management*.

5.1.8 Differentiated Gap-assisted Measurement

With gap-assisted measurement, the gNodeB configures periodic idle time for RRC_CONNECTED UEs, during which these UEs measure cell signal quality on specified frequencies. However, scheduling is suspended during these measurements, affecting service experience of UEs. Continuous gap-assisted measurement will be triggered in the following scenarios:

- Inter-frequency or inter-RAT periodic reporting of measurement reports (MRs). For details, see section 5.5 "Measurements" in 3GPP TS 38.331 V15.5.0.
- Fast automatic neighbor relation (ANR) or PCI-specific ANR. For details, see *ANR*.
- Proactive PCI conflict detection based on intra-RAT ANR. For details, see *PCI Conflict Detection and Self-Optimization*. PCI stands for physical cell identifier.

Differentiated gap-assisted measurement is introduced to reduce the impact of gap-assisted measurement on UEs with QCI-specific bearers. After this function is enabled, **UEs with QCI-specific bearers do not participate in gap-assisted measurement, increasing the scheduling probability of these UEs and shortening the scheduling delay.** This function includes the following optimizations.

Optimization	Description	Function Switch	Setting Notes
Exit from gap-assisted measurement for UEs with QCI-specific bearers	If UEs with QCI-specific bearers have been selected by the gNodeB for gap-assisted measurements, the gNodeB delivers an RRC Connection Reconfiguration message to delete the corresponding measurement procedure, thereby avoiding such measurements.	**SA_EXIT_FROM_MR_GAP_MEAS_SW** option of the **NrCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter	Select this option.
Prohibition of gap-assisted measurement for UEs with QCI-specific bearers after an exit from such measurements	If periodic inter-frequency or inter-RAT MR events (including management-based MDT events) have been subscribed to, this optimization can be used to prevent UEs with QCI-specific bearers from performing gap-assisted measurement.	**SA_ENTER_INTO_MR_GAP_MEAS_SW** option of the **NrCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter	Deselect this option.
Exclusion of UEs with QCI-specific bearers from periodic inter-frequency or inter-RAT MR event reporting	If periodic inter-frequency or inter-RAT MR events (including management-based MDT events) have been subscribed to, this optimization can be used to exclude UEs with QCI-specific bearers from periodic inter-frequency or inter-RAT MR event reporting.	**SA_INTER_FREQ_RAT_MR_SELECT_SW** option of the **gNBQciBearer**.***ServiceDiffAlgoSwitch*** parameter	Deselect this option.

NOTE

When multiple bearers are set up for a UE:

- If the **SA_EXIT_FROM_MR_GAP_MEAS_SW** option of the **NrCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter is selected for any bearer of the UE with gap-assisted measurement in effect, the UE will exit gap-assisted measurement.
- If the **SA_ENTER_INTO_MR_GAP_MEAS_SW** option of the **NrCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter is deselected for any bearer of the UE which has exited gap-assisted measurement, the UE will not be allowed to perform gap-assisted measurement.
- If the **SA_INTER_FREQ_RAT_MR_SELECT_SW** option of the **gNBQciBearer**.***ServiceDiffAlgoSwitch*** parameter is deselected for any bearer, the corresponding UEs are excluded from periodic inter-frequency or inter-RAT MR event reporting.

5.1.9 Explicit Scheduling for Downlink Second Retransmission

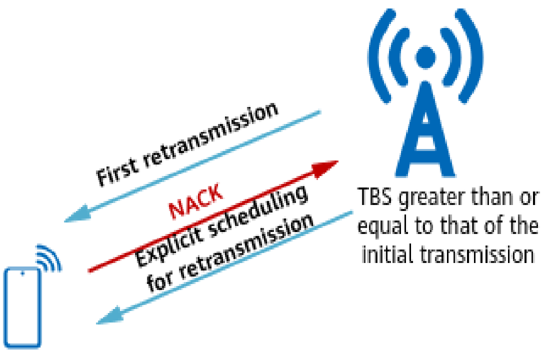
Currently, there are two methods of instructing the TBS used for downlink HARQ retransmissions: implicit scheduling and explicit scheduling.

- Implicit scheduling for HARQ retransmissions: The DCI used for retransmissions does not indicate a specific MCS index. This method is commonly used when a UE sends a NACK to trigger scheduling for retransmission. As the UE has received downlink data in this scenario, the UE knows the TBS of the initially transmitted data, which must remain unchanged for retransmission. Therefore, the UE can calculate the MCS index based on the TBS and the number of RBs.
- Explicit scheduling for HARQ retransmissions: The DCI used for retransmissions indicates a specific MCS index. This method is commonly used when a UE sends discontinuous transmission (DTX) information to trigger retransmission. As the UE does not receive downlink data in this scenario, the UE does not know the TBS and cannot calculate the MCS index. Therefore, the DCI used for retransmission needs to indicate a specific MCS to the UE.

If the UE does not receive the DCI result for PDSCH scheduling, it sends DTX information to the gNodeB. If the gNodeB mistakenly considers a DTX as a NACK, implicit scheduling for downlink HARQ retransmissions is used. In this case, consecutive retransmission failures may occur, causing packet losses. To avoid this problem and retain the combining gains of the first retransmission, explicit scheduling for downlink second retransmission is introduced.

This function optimizes the second retransmission mechanism for UEs with QCI-specific bearers when the base station detects that the first retransmission result reported by the UEs is NACK. As shown in **Figure 5-5**, after this function is enabled, if the TBS allowed at the second retransmission is greater than or equal to the TBS allowed at the initial transmission, the base station uses explicit scheduling for retransmission. That is, the base station sends the DCI used for retransmission to the UEs to indicate a specific MCS index. This increases the demodulation success rate of the UEs and therefore improves the downlink transmission reliability. This function is controlled by the **DL_SECOND_RETRANS_EXPLICIT_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter.

Figure 5-5 Explicit scheduling for downlink second retransmission



NOTE

When multiple bearers are set up for a UE:

- If data of multiple bearers needs to be scheduled at the scheduling moment, the setting of the **DL_SECOND_RETRANS_EXPLICIT_SW** option for the QCI with the highest priority takes effect for the UE. For details about QCI configurations of UEs and QCI priorities, see *QoS Management*.
- If only data of a single bearer is to be scheduled at the scheduling moment, the setting of the **DL_SECOND_RETRANS_EXPLICIT_SW** option for this bearer takes effect.

Explicit scheduling for downlink second retransmission is also supported in Cloud X service scenarios. For details, see *Ultimate Cloud X Experience*.

5.2 Network Analysis

5.2.1 Benefits

Deployment Suggestions

It is recommended that a 5QI or QCI be planned separately for a service that requires differentiated guarantee and that the separately-planned 5QI or QCI not be used for other services. On this basis, service-differentiated link reliability functions are enabled for the service's QCI.

It is recommended that DRX and power saving BWP be disabled for specific QCIs. This avoids affecting the delay of specific services. For details about DRX, see *DRX*. For details about power saving BWP, see *UE Power Saving*.

Benefits

Function	Benefit Analysis
QCI-specific PDCCH aggregation level optimization	With this function, the PDCCH aggregation level of UEs with QCI-specific bearers increases and the PDCCH BLER decreases.
QCI-specific uplink MCS selection optimization	With this function, the uplink IBLER of UEs with QCI-specific bearers decreases, the 5QI/QCI-specific average uplink delay at the MAC layer for QCI-specific bearers decreases, and the number of used uplink RBs increases. A larger value set for the **gNBDUSchParamGroup**.***UlInitTransMcsDecrease*** parameter results in a greater decrease in the uplink IBLER of UEs with QCI-specific bearers, a greater decrease in the 5QI/QCI-specific average uplink delay at the MAC layer for QCI-specific bearers, and a greater increase in the number of used uplink RBs. However, if the **gNBDUSchParamGroup**.***UlInitTransMcsDecrease*** parameter is set to an excessively large value, the spectral efficiency of uplink transmission decreases significantly and the transmission duration is prolonged.

Function	Benefit Analysis
QCI-specific downlink MCS selection optimization	With this function, the downlink IBLER of UEs with QCI-specific bearers decreases, the 5QI/QCI-specific average downlink delay at the MAC layer for QCI-specific bearers decreases, and the number of used downlink RBs increases. A larger value set for the **gNBDUSchParamGroup**.***DlCqiAdjValue*** parameter results in a greater decrease in the downlink IBLER of UEs with QCI-specific bearers, a greater decrease in the 5QI/QCI-specific average downlink delay at the MAC layer for QCI-specific bearers, and a greater increase in the number of used downlink RBs. However, if the **gNBDUSchParamGroup**.***DlCqiAdjValue*** parameter is set to an excessively large value, the spectral efficiency of downlink transmission decreases significantly and the transmission duration is prolonged.
QCI-specific uplink MU pairing differentiation	With this function, the uplink MU-MIMO pairing ratio of UEs with QCI-specific bearers and the uplink IBLER decrease.
QCI-specific downlink MU pairing assurance	With this function, the downlink MU-MIMO pairing ratio of UEs with QCI-specific bearers and the downlink IBLER decrease.
QCI-specific downlink power control optimization	With this function, the downlink power per RB increases for UEs with QCI-specific bearers and the downlink IBLER and downlink retransmission proportion may decrease.
QCI-specific uplink power control optimization	With this function, the uplink power per RB increases for UEs with QCI-specific bearers and the uplink IBLER and uplink retransmission proportion may decrease.
Differentiated gap-assisted measurement	With this function enabled in scenarios with continuous gap-assisted measurements, UEs will exit and no longer start gap-assisted measurements. As a result, the scheduling opportunities increase, and the 5QI/QCI-specific average uplink or downlink delay at the MAC layer for QCI-specific bearers decreases.
Explicit scheduling for downlink second retransmission	With this function, the 5QI/QCI-specific average downlink delay at the MAC layer for QCI-specific bearers decreases.
For details about related network indicators, see **Table 5-1**.	

Table 5-1 Definitions of indicators related to benefit analysis of service-differentiated link reliability

Indicator Name	Indicator Definition
Uplink MU-MIMO pairing ratio	**N.ChMeas.MIMO.UL.Pair.Ratio**
Downlink MU-MIMO pairing ratio	**N.ChMeas.MIMO.DL.Pair.Ratio**
Uplink IBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100%
Downlink IBLER	(**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100%
Downlink retransmission proportion	(**N.DL.SCH.QPSK.TB.Retrans** + **N.DL.SCH.16QAM.TB.Retrans** + **N.DL.SCH.64QAM.TB.Retrans** + **N.DL.SCH.256QAM.TB.Retrans**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB** + **N.DL.SCH.QPSK.TB.Retrans** + **N.DL.SCH.16QAM.TB.Retrans** + **N.DL.SCH.64QAM.TB.Retrans** + **N.DL.SCH.256QAM.TB.Retrans**) x 100%
Uplink retransmission proportion	(**N.UL.SCH.HalfPiBPSK.TB.Retrans** + **N.UL.SCH.QPSK.TB.Retrans** + **N.UL.SCH.16QAM.TB.Retrans** + **N.UL.SCH.64QAM.TB.Retrans** + **N.UL.SCH.256QAM.TB.Retrans**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB** + **N.UL.SCH.HalfPiBPSK.TB.Retrans** + **N.UL.SCH.QPSK.TB.Retrans** + **N.UL.SCH.16QAM.TB.Retrans** + **N.UL.SCH.64QAM.TB.Retrans** + **N.UL.SCH.256QAM.TB.Retrans**) x 100%
5QI-specific average uplink delay at the MAC layer	**N.MAC.UL.Cell5QI.Delay**/**N.MAC.UL.Packets.Cell5QI**

Indicator Name	Indicator Definition
5QI-specific average downlink delay at the MAC layer	**N.MAC.DL.Cell5QI.Delay/N.MAC.DL.Packets.Cell5QI**
QCI-specific average uplink delay at the MAC layer	**N.MAC.UL.QCI.Delay/N.MAC.UL.Packets.QCI**
QCI-specific average downlink delay at the MAC layer	**N.MAC.DL.QCI.Delay/N.MAC.DL.Packets.QCI**

5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

Function	Network Impact
QCI-specific PDCCH aggregation level optimization	With this function, the network impact is related to the uplink and downlink traffic volumes of UEs with QCI-specific bearers. A larger number of UEs with QCI-specific bearers results in a higher average PDCCH aggregation level and lower uplink and downlink CCE allocation success rates in a cell.
QCI-specific uplink MCS selection optimization	With this function, the QCI-specific uplink data rate (in NSA networking) or 5QI-specific uplink data rate (in SA networking) decreases. The network impact is related to the uplink traffic volume of UEs with QCI-specific bearers. A larger number of UEs with QCI-specific bearers brings the following impacts to a cell: (1) higher uplink RB usage; (2) lower average uplink MCS index, uplink BLER, and uplink user-perceived rate; (3) possibly higher 5QI-specific average uplink delay at the RLC layer for QCI-specific bearers. In high-load scenarios, the average uplink cell throughput decreases.

Function	Network Impact
QCI-specific downlink MCS selection optimization	With this function, the QCI-specific downlink data rate (in NSA networking) or 5QI-specific downlink data rate (in SA networking) decreases. The network impact is related to the downlink traffic volume of UEs with QCI-specific bearers. A larger number of UEs with QCI-specific bearers brings the following impacts to a cell: (1) higher downlink RB usage; (2) lower average downlink MCS index, downlink BLER, and downlink user-perceived rate; (3) possibly higher 5QI-specific average downlink delay at the RLC layer for QCI-specific bearers. In high-load scenarios, the average downlink cell throughput decreases.
QCI-specific uplink MU pairing differentiation	With this function, the network impact is related to the uplink traffic volume of UEs with QCI-specific bearers. A larger number of UEs with QCI-specific bearers results in higher uplink RB usage and lower uplink MU-MIMO pairing ratio and uplink user-perceived rate in a cell. In high-load scenarios, the average uplink cell throughput decreases.
QCI-specific downlink MU pairing assurance	With this function, the network impact is related to the downlink traffic volume of UEs with QCI-specific bearers. A larger number of UEs with QCI-specific bearers results in higher downlink RB usage and lower downlink MU-MIMO pairing ratio and downlink user-perceived rate in a cell. In high-load scenarios, the average downlink cell throughput decreases.
QCI-specific downlink power control optimization	With this function, the downlink transmit power of UEs with QCI-specific bearers increases. As a result, the downlink interference to neighboring cells increases and the average downlink MCS index of neighboring cells decreases.
QCI-specific uplink power control optimization	With this function, the uplink transmit power of UEs with QCI-specific bearers increases. As a result, the uplink interference to neighboring cells increases and the average uplink MCS index of neighboring cells decreases.
Differentiated gap-assisted measurement	With this function in scenarios where gap-assisted measurement is performed, UEs do not start or continue with gap-assisted measurement and the scheduling probability increases. As a result, the number of RRC connection reconfiguration messages over the air interface increases, thereby increasing the service drop rate.
Explicit scheduling for downlink second retransmission	None
For details about related network indicators, see **Table 5-2**.	

Table 5-2 Definitions of indicators related to network analysis of service-differentiated link reliability

Indicator Name	Indicator Definition
Uplink CCE allocation success rate	(**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)/**N.CCE.UL.AllocReq.Num**
Downlink CCE allocation success rate	(**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)/**N.CCE.DL.AllocReq.Num**
Average uplink MCS index	SUM(***k*** x **N.ChMeas.PUSCH.MCS.*k***)/ SUM(**N.ChMeas.PUSCH.MCS.*k***), where ***k*** ranges from **0** to **31**
Average downlink MCS index	SUM(***k*** x **N.ChMeas.PDSCH.MCS.*k***)/ SUM(**N.ChMeas.PDSCH.MCS.*k***), where ***k*** ranges from **0** to **28**
QCI-specific uplink data rate	(**N.ThpVolForRate.UL.QCI – N.ThpVolForRate.UL.SmallPkt.QCI**)/ **N.ThpTimeForRate.UL.RmvSmallPkt.QCI**
QCI-specific downlink data rate	(**N.ThpVol.DL.QCI – N.ThpVol.DL.LastSlot.QCI**)/ **N.ThpTime.DL.RmvLastSlot.QCI**
5QI-specific uplink data rate	(**N.ThpVolForRate.UL.DuCell5QI – N.ThpVolForRate.UL.SmallPkt.DuCell5QI**)/ **N.ThpTimeForRate.UL.RmvSmallPkt.DuCell5QI**
5QI-specific downlink data rate	(**N.ThpVol.DL.DuCell5QI – N.ThpVol.DL.LastSlot.DuCell5QI**)/ **N.ThpTime.DL.RmvLastSlot.DuCell5QI**
Uplink RB usage	**Uplink Resource Block Utilizing Rate (DU)**
Downlink RB usage	**Downlink Resource Block Utilizing Rate (DU)**
Uplink IBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100%
Downlink IBLER	(**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100%

Indicator Name	Indicator Definition
Average uplink cell throughput	**Cell Uplink Average Throughput (DU)**
Average downlink cell throughput	**Cell Downlink Average Throughput (DU)**
5QI-specific average uplink delay at the RLC layer	**N.RLC.UL.Cell5QI.Delay**/**N.RLC.UL.Packets.Cell5QI**
5QI-specific average downlink delay at the RLC layer	**N.RLC.DL.Cell5QI.Delay**/ **N.RLC.DL.TrfSDU.RxPackets.DuCell5QI**
Uplink MU-MIMO pairing ratio	**N.ChMeas.MIMO.UL.Pair.Ratio**
Downlink MU-MIMO pairing ratio	**N.ChMeas.MIMO.DL.Pair.Ratio**
Service drop rate	**Service Call Drop Rate (CU)**

5.3 Requirements

5.3.1 Licenses

The following table lists the license required for each sub-function of service-differentiated link reliability (excluding QCI-specific uplink power control optimization, QCI-specific uplink MU pairing differentiation, and differentiated gap-assisted measurement).

Feature ID	Feature Name	Model	Sales Unit
FOFD-091202	Game-like Service Latency Assurance	NR0S00LSG000	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

QCI-specific uplink power control optimization, QCI-specific uplink MU pairing differentiation, and differentiated gap-assisted measurement: none

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

QCI-specific uplink MU pairing differentiation

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH MU-MIMO	**UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	QCI-specific uplink MU pairing differentiation can take effect only when the **UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter is selected.

QCI-specific downlink MU pairing assurance

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink MU-MIMO	**DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	QCI-specific downlink MU pairing assurance can take effect only when the **DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter is selected.

Other sub-functions: None

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

QCI-specific uplink MCS selection optimization

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Optimized uplink MCS index selection	**VONR_DMRS_SCH_SW** option of the **NRDUCellUlAmc.*UlAmcPerformanceSw*** parameter	*VoNR*	When both optimized uplink MCS index selection for VoNR and QCI-specific uplink MCS selection optimization are enabled, the decrease in the MCS index for uplink initial transmission of VoNR UEs is specified by the larger of either the **NRDUCellUlAmc.*VonrUlInitTransMcsDe-crease*** or **gNBDUSchParamGroup**.***UlInitTransMcsDe-crease*** parameter value.

QCI-specific downlink MCS selection optimization

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast MCS index reduction in the downlink	**NRDUCellDlAmc.*VoiceDlCqiAdjValue***	*VoNR*	When both fast MCS index reduction in the downlink for VoNR and QCI-specific downlink MCS selection optimization are enabled, the downlink CQI adjustment value specified by the **NRDUCellDlAmc.*VoiceDlCqiAdjValue*** parameter takes effect for VoNR UEs.

QCI-specific downlink power control optimization

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power aggregation in unoccupied bandwidths	**NRDUCellChnPwr**.***MaxPdschConvPwrOffset***	*Power Control*	QCI-specific downlink power control optimization does not take effect if the **NRDUCellChnPwr**.***MaxPdschConvPwrOff-set*** parameter is set to a non-zero value. The upper limit of the actual increase in the PDSCH power per RB depends on the **NRDUCellChnPwr**.***MaxPdschConvPwrOff-set*** parameter value.
Low-frequency TDD	DL 256QAM	**NRDUCellAlgoSwitch**.***Dl256QamSwitch***	*Modulation Schemes*	If there are UEs for which DL 256QAM is being used, QCI-specific downlink power control optimization may take effect for a lower proportion of UEs with QCI-specific bearers.
Low-frequency TDD	DL 1024QAM	**DL_1024QAM_SW** option of the **NRDUCellDlAmc**.***DlModulationSchemeSw*** parameter	*Modulation Schemes*	If there are UEs for which DL 1024QAM is being used, QCI-specific downlink power control optimization may take effect for a lower proportion of UEs with QCI-specific bearers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink power compensation	**DL_SERV_EXP_IMP_SW** option of the **gNBDUQciAlgoParamGrp.*QciAlgoSwitch*** parameter	*VoNR*	Whether downlink power aggregation takes effect for VoNR UEs that have multiple bearers depends on the setting of the **DL_SERV_EXP_IMP_SW** option of the **gNBDUQciAlgoParamGrp.*QciAlgoSwitch*** parameter corresponding to the voice bearers of these UEs. Downlink power aggregation takes effect for VoNR UEs if the **DL_SERV_EXP_IMP_SW** option of the **gNBDUQciAlgoParamGrp.*QciAlgoSwitch*** parameter is selected for the voice bearers of these UEs.

Other sub-functions: None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-3 describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 5-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
QCI-based Diff Service Experience Switch	**NRDUCellFeatureSw**.***QciBasedDiffServExpSw***	LINK_RELIABILITY_DIFF_SW	Select this option if service-differentiated link reliability is required.
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	PUSCH_DTX_BASED_PDCCH_ADJ_SW	Select this option if QCI-specific PDCCH aggregation level optimization is required.
PDCCH SINR Offset	**gNBDUSchParamGroup**.***PdcchSinrOffset***	None	Set this parameter to **30**.
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	DL_BLER_TAR_QCI_SW	Select this option.
PDCCH BLER Target	**gNBDUSchParamGroup**.***PdcchBlerTarget***	None	Set this parameter to **2**.

Parameter Name	Parameter ID	Option	Setting Notes
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	SCH_MCS_SELECT_OPT_SW	Select this option if QCI-specific uplink MCS selection optimization or QCI-specific downlink MCS selection optimization is required.
UL Initial Transmission MCS Decrease	**gNBDUSchParamGroup**.***UlInitTransMcsDecrease***	None	Set this parameter based on the network plan after QCI-specific uplink MCS selection optimization is enabled.
Downlink CQI Adjust Value	**gNBDUSchParamGroup**.***DlCqiAdjValue***	None	Set this parameter based on the network plan after QCI-specific downlink MCS selection optimization is enabled.
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	DL_PWR_AGG_QCI_SW	Select this option if QCI-specific downlink power control optimization is required.
Service Differentiation Algorithm Switch	**NRDUCellQciBearer**.***ServiceDiffAlgoSwitch***	UL_MU_MIMO_SW	Deselect this option if QCI-specific uplink MU pairing differentiation is required.
Service Differentiation Algorithm Switch	**NRDUCellQciBearer**.***ServiceDiffAlgoSwitch***	DL_MU_MIMO_CORR_THLD_ADJ_SW	Select this option if QCI-specific downlink MU pairing assurance is required.
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	UL_PWR_CTRL_CONFIG_QCI_SW	Select this option if QCI-specific uplink power control optimization is required.
Near Point PUSCH SINR Threshold Offset	**gNBDUSchParamGroup**.***NearPointPuschSinrThldOfs***	None	Set this parameter to **0**.
PUSCH RSRP Threshold	**gNBDUSchParamGroup**.***PuschRsrpThld***	None	Set this parameter to **-71**.

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Outer-Loop Power Control Target	**gNBDUSchParamGroup**.***PuschOuterLoopPcTarget***	None	Set this parameter to **4**.
Service Differentiation Algorithm Switch	**NrCellQciBearer**.***ServiceDiffAlgoSwitch***	SA_EXIT_FROM_MR_GAP_MEAS_SW	Select this option.
Service Differentiation Algorithm Switch	**NrCellQciBearer**.***ServiceDiffAlgoSwitch***	SA_ENTER_INTO_MR_GAP_MEAS_SW	Deselect this option.
Service Differentiation Algorithm Switch	**gNBQciBearer**.***ServiceDiffAlgoSwitch***	SA_INTER_FREQ_RAT_MR_SELECT_SW	Deselect this option.
Service Differentiation Algorithm Switch	**NRDUCellQciBearer**.***ServiceDiffAlgoSwitch***	DL_SECOND_RETRANS_EXPLICIT_SW	Select this option.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

This function takes effect 5 to 10 minutes after the related parameter configurations in the **gNBDUSchParamGroup** and **NRDUCellQciBearer** MOs are modified.

Activation Command Examples

- Basic configuration for service-differentiated link reliability

```
//Turning on the switch for service-differentiated link reliability
MOD NRDUCELLFEATURESW: NrDuCellId=1, QciBasedDiffServExpSw=LINK_RELIABILITY_DIFF_SW-1;
//Adding a gNodeB DU scheduling parameter group
```

```
ADD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1;
//Mapping the gNodeB DU scheduling parameter group to a bearer of a specific QCI
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, ScheduleParamGroupId=1;
```

- QCI-specific PDCCH aggregation level optimization

```
//Turning on the PUSCH-DTX-based PDCCH Adjustment Switch
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=PUSCH_DTX_BASED_PDCCH_ADJ_SW-1;
//Turning on the QCI-based PDCCH BLER Target Switch and setting the target PDCCH BLER and
PDCCH SINR offset
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=DL_BLER_TAR_QCI_SW-1;
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, PdcchBlerTarget=2;
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, PdcchSinrOffset=30;
```

- QCI-specific uplink MCS selection optimization

```
//Turning on the switch for QCI-specific uplink MCS selection optimization
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=SCH_MCS_SELECT_OPT_SW-1;
//Modifying the QCI-specific decrease in the MCS index for uplink initial transmission
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,UlInitTransMcsDecrease=1;
```

- QCI-specific downlink MCS selection optimization

```
//Turning on the switch for QCI-specific downlink MCS selection optimization
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=SCH_MCS_SELECT_OPT_SW-1;
//Modifying the QCI-specific downlink CQI adjustment value
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, DlCqiAdjValue=10;
```

- QCI-specific uplink MU pairing differentiation

```
//Turning off the uplink MU-MIMO switch
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, ServiceDiffAlgoSwitch=UL_MU_MIMO_SW-0;
```

- QCI-specific downlink MU pairing assurance

```
//Turning on the switch for QCI-specific downlink MU pairing assurance
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9,
ServiceDiffAlgoSwitch=DL_MU_MIMO_CORR_THLD_ADJ_SW-1;
```

- QCI-specific downlink power control optimization

```
//Turning on the switch for QCI-specific downlink power control optimization
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=DL_PWR_AGG_QCI_SW-1;
```

- QCI-specific uplink power control optimization

```
//Turning on the switch for QCI-specific uplink power control optimization
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_PWR_CTRL_CONFIG_QCI_SW-1;
//Modifying the PUSCH SINR threshold offset for UEs located at the cell center
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, NearPointPuschSinrThldOfs=0;
//Modifying the PUSCH RSRP threshold
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, PuschRsrpThld=-71;
//Modifying the target value for PUSCH outer-loop power control
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, PuschOuterLoopPcTarget=4;
```

- Differentiated gap-assisted measurement

```
//Turning on the SA Exit from Gap-assisted Measure Sw
MOD NRCELLQCIBEARER: NrCellId=1, Qci=9,
ServiceDiffAlgoSwitch=SA_EXIT_FROM_MR_GAP_MEAS_SW-1;
//Turning off the SA Entry into Gap-assisted Measure Sw
MOD NRCELLQCIBEARER: NrCellId=1, Qci=9,
ServiceDiffAlgoSwitch=SA_ENTER_INTO_MR_GAP_MEAS_SW-0;
//Turning off the SA Inter-Freq Inter-RAT Prdc MR Sel Sw
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=SA_INTER_FREQ_RAT_MR_SELECT_SW-0;
```

- Explicit scheduling for downlink second retransmission

```
//Turning on the switch for explicit scheduling for downlink second retransmission
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9,
ServiceDiffAlgoSwitch=DL_SECOND_RETRANS_EXPLICIT_SW-1;
```


Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Differentiated gap-assisted measurement
```
//Turning off the SA Exit from Gap-assisted Measure Sw
MOD NRCELLQCIBEARER: NrCellId=1, Qci=9,
ServiceDiffAlgoSwitch=SA_EXIT_FROM_MR_GAP_MEAS_SW-0;
//Turning on the SA Entry into Gap-assisted Measure Sw
MOD NRCELLQCIBEARER: NrCellId=1, Qci=9,
ServiceDiffAlgoSwitch=SA_ENTER_INTO_MR_GAP_MEAS_SW-1;
//Turning on the SA Inter-Freq Inter-RAT Prdc MR Sel Sw
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=SA_INTER_FREQ_RAT_MR_SELECT_SW-1;
```
- Explicit scheduling for downlink second retransmission
```
//Turning off the switch for explicit scheduling for downlink second retransmission
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9,
ServiceDiffAlgoSwitch=DL_SECOND_RETRANS_EXPLICIT_SW-0;
```
- QCI-specific PDCCH aggregation level optimization
```
//Turning off the PUSCH-DTX-based PDCCH Adjustment Switch
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=PUSCH_DTX_BASED_PDCCH_ADJ_SW-0;
//Turning off the QCI-based PDCCH BLER Target Switch
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=DL_BLER_TAR_QCI_SW-0;
```
- QCI-specific uplink and downlink MCS selection optimization
```
//Turning off the switch for MCS selection optimization for QCI-based scheduling
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=SCH_MCS_SELECT_OPT_SW-0;
```
- QCI-specific uplink MU pairing differentiation
```
//Turning on the uplink MU-MIMO switch
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, ServiceDiffAlgoSwitch=UL_MU_MIMO_SW-1;
```
- QCI-specific downlink MU pairing assurance
```
//Turning off the switch for QCI-specific downlink MU pairing assurance
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9,
ServiceDiffAlgoSwitch=DL_MU_MIMO_CORR_THLD_ADJ_SW-0;
```
- QCI-specific downlink power control optimization
```
//Turning off the switch for QCI-specific downlink power control optimization
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=DL_PWR_AGG_QCI_SW-0;
```
- QCI-specific uplink power control optimization
```
//Turning off the switch for QCI-specific uplink power control optimization
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_PWR_CTRL_CONFIG_QCI_SW-0;
```
- Basic configuration for service-differentiated link reliability
```
//Removing the mapping between the gNodeB DU scheduling parameter group and the bearer of a specific QCI
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, ScheduleParamGroupId=255;
//Removing the gNodeB DU scheduling parameter group
RMV GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1;
//Turning off the switch for service-differentiated link reliability
MOD NRDUCELLFEATURESW: NrDuCellId=1, QciBasedDiffServExpSw=LINK_RELIABILITY_DIFF_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

QCI-specific PDCCH Aggregation Level Optimization

On the MAE-Access, perform the following steps to check whether QCI-specific PDCCH aggregation level optimization has taken effect:

Step 1 Enable a UE to access a cell. After a bearer with a specific QCI or 5QI is set up, enable the UE to perform services.

Step 2 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 3 In the left navigation pane of the displayed **Signaling Trace Management** tab page, choose **User Performance Monitoring** > **BLER Monitoring** to observe the PDCCH BLER. If the values of related counters decrease, the function has taken effect.

----End

QCI-specific Uplink MCS Selection Optimization, QCI-specific Uplink MU Pairing Differentiation, and QCI-specific Uplink Power Control Optimization

On the MAE-Access, perform the following steps to check whether QCI-specific uplink MCS selection optimization, QCI-specific uplink MU pairing differentiation, or QCI-specific uplink power control optimization has taken effect:

Step 1 Enable a UE to access a cell. After a bearer with a specific QCI or 5QI is set up, enable the UE to perform services.

Step 2 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 3 In the left navigation pane of the displayed **Signaling Trace Management** tab page, choose **User Performance Monitoring** > **BLER Monitoring** to observe the uplink IBLER. If the values of related counters decrease, the function has taken effect.

----End

QCI-specific Downlink MCS Selection Optimization and QCI-specific Downlink Power Control Optimization

On the MAE-Access, perform the following steps to check whether QCI-specific downlink MCS selection optimization or QCI-specific downlink power control optimization has taken effect:

Step 1 Enable a UE to access a cell. After a bearer with a specific QCI or 5QI is set up, enable the UE to perform services.

Step 2 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 3 In the left navigation pane of the displayed **Signaling Trace Management** tab page, choose **User Performance Monitoring** > **BLER Monitoring** to observe the downlink IBLER. If the values of related counters decrease, the function has taken effect.

----End

QCI-specific Downlink MU Pairing Assurance

On the MAE-Access, perform the following steps to check whether QCI-specific downlink MU pairing assurance has taken effect:

Step 1 Enable a UE to access a cell. After a bearer with a specific QCI or 5QI is set up, enable the UE to perform services.

Step 2 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 3 In the left navigation pane of the displayed **Signaling Trace Management** tab page, choose **User Performance Monitoring** > **User Common Monitoring** to observe the downlink MU scheduling times. If the number of downlink MU pairing times decreases, the function has taken effect.

----End

Differentiated Gap-assisted Measurement

1. Run the following MML commands to subscribe to the performance counter objects.
   - SA networking (using 5QI 9 as an example)
     ```
     MOD GNB5QICONFIG: Nr5qi=9, CounterObjSubscriptionInd=YES;
     ```
   - NSA networking (using QCI 9 as an example)
     ```
     MOD GNBQCIBEARER: Qci=9, CounterObjSubscriptionInd=YES;
     ```
2. In scenarios with gap-assisted measurement, after differentiated gap-assisted measurement is enabled, use the following methods to observe whether this function has taken effect.
   - In SA networking, if the values of the following indicators decrease, this function has taken effect.

Indicator Name	Definition
5QI-specific average uplink delay at the MAC layer	**N.MAC.UL.Cell5QI.Delay/N.MAC.UL.Packets.Cell5QI**
5QI-specific average downlink delay at the MAC layer	**N.MAC.DL.Cell5QI.Delay/N.MAC.DL.Packets.Cell5QI**

– In NSA networking, if the values of the following indicators decrease, this function has taken effect.

Indicator Name	Definition
QCI-specific average uplink delay at the MAC layer	**N.MAC.UL.QCI.Delay/N.MAC.UL.Packets.QCI**
QCI-specific average downlink delay at the MAC layer	**N.MAC.DL.QCI.Delay/N.MAC.DL.Packets.QCI**

Explicit Scheduling for Downlink Second Retransmission

1. Run the following MML commands to subscribe to the performance counter objects.
   – SA networking (using 5QI 9 as an example)
   ```
   MOD GNB5QICONFIG: Nr5qi=9, CounterObjSubscriptionInd=YES;
   ```
   – NSA networking (using QCI 9 as an example)
   ```
   MOD GNBQCIBEARER: Qci=9, CounterObjSubscriptionInd=YES;
   ```
2. After explicit scheduling for downlink second retransmission is enabled, use the following methods to observe whether this function has taken effect.
   – In SA networking, if the value of the following indicator decreases, this function has taken effect.

Indicator Name	Definition
5QI-specific average downlink delay at the MAC layer	**N.MAC.DL.Cell5QI.Delay/N.MAC.DL.Packets.Cell5QI**

   – In NSA networking, if the value of the following indicator decreases, this function has taken effect.

Indicator Name	Definition
QCI-specific average downlink delay at the MAC layer	**N.MAC.DL.QCI.Delay/N.MAC.DL.Packets.QCI**

5.4.3 Network Monitoring

The indicators listed in **5.2 Network Analysis** can be used to monitor the benefits and impacts of this function.

6 QCI-specific CCE Aggregation Level Adaptation

6.1 Principles

"QCI-specific PDCCH aggregation level optimization" increases the PDCCH CCE aggregation level for UEs with QCI-specific bearers, thereby reducing the probability of DCI miss-detection and improving the experience of these UEs. For details, see **5.1.1 QCI-specific PDCCH Aggregation Level Optimization**. However, enabling "QCI-specific PDCCH aggregation level optimization" will exacerbate the insufficiency of CCE resources in cells with a high CCE allocation failure rate. As a result, the scheduling opportunities of UEs with QCI-specific bearers and other UEs in the cell are reduced, thereby affecting user experience. To address this issue, QCI-specific CCE aggregation level adaptation is introduced.

After "QCI-specific CCE aggregation level adaptation" is enabled together with "QCI-specific PDCCH aggregation level optimization", the following rules apply to UEs with QCI-specific bearers:

- When the uplink or downlink CCE allocation failure rate of a cell exceeds the upper threshold specified by the **gNBDUSchParamGroup**.***CceAllocFailRateUpperThld*** parameter, **"QCI-specific PDCCH aggregation level optimization" cannot take effect for UEs with QCI-specific bearers. This prevents the scheduling opportunities of UEs with QCI-specific bearers and other UEs in the cell from being affected, thereby improving their experience.**
- When both the uplink and downlink CCE allocation failure rates in the cell fall below the lower threshold specified by the **gNBDUSchParamGroup**.***CceAllocFailRateLowerThld*** parameter, "QCI-specific PDCCH aggregation level optimization" can take effect for UEs with QCI-specific bearers. This reduces the probability of DCI miss-detection for these UEs, thereby improving their experience.

QCI-specific CCE aggregation level adaptation is controlled by the **CCE_AGG_LEVEL_ADAPT_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter.

NOTE

When multiple bearers are set up for a UE, the function parameter settings corresponding to the UE's QCI with the highest priority take effect. For details about QCI configurations of UEs and QCI priorities, see *QoS Management*.

6.2 Network Analysis

6.2.1 Benefits

Assume that QCI-specific CCE aggregation level adaptation is enabled together with any optimization of "QCI-specific PDCCH aggregation level optimization" in scenarios with a high CCE allocation failure rate in a cell. In this situation, the cell's uplink and downlink CCE allocation success rates increase, thereby increasing the values of **Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**.

Table 6-1 Definitions of indicators related to benefits

Indicator Name	Definition
Uplink CCE allocation success rate	(**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)/**N.CCE.UL.AllocReq.Num**
Downlink CCE allocation success rate	(**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)/**N.CCE.DL.AllocReq.Num**

6.2.2 Impacts

The impacts between this function and other functions are described in **6.3.2.3 Function Impacts**.

Assume that QCI-specific CCE aggregation level adaptation is enabled together with any optimization of "QCI-specific PDCCH aggregation level optimization" in scenarios with a high CCE allocation failure rate in a cell. In this situation, the average uplink and downlink PDCCH CCE aggregation levels of UEs with QCI-specific bearers decrease, which increases the uplink and downlink CCE allocation success rates of the cell and possibly increases the DCI miss-detection rate of such UEs. As a result, the uplink and downlink rates of a specific QCI or 5QI, and 5QI-specific average uplink and downlink delays at the MAC layer fluctuate.

Table 6-2 Definitions of indicators related to impacts

Indicator Name	Definition
Average uplink PDCCH CCE aggregation level	(**N.CCE.UL.AggLvl16Num** x 16 + **N.CCE.UL.AggLvl8Num** x 8 + **N.CCE.UL.AggLvl4Num** x 4 + **N.CCE.UL.AggLvl2Num** x 2 + **N.CCE.UL.AggLvl1Num**)/(**N.CCE.UL.AggLvl16Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl1Num**)
Average downlink PDCCH CCE aggregation level	(**N.CCE.DL.AggLvl16Num** x 16 + **N.CCE.DL.AggLvl8Num** x 8 + **N.CCE.DL.AggLvl4Num** x 4 + **N.CCE.DL.AggLvl2Num** x 2 + **N.CCE.DL.AggLvl1Num**)/(**N.CCE.DL.AggLvl16Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl1Num**)
Uplink CCE allocation success rate	(**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)/**N.CCE.UL.AllocReq.Num**
Downlink CCE allocation success rate	(**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)/**N.CCE.DL.AllocReq.Num**
QCI-specific downlink data rate	(**N.ThpVol.DL.QCI-N.ThpVol.DL.LastSlot.QCI**)/ **N.ThpTime.DL.RmvLastSlot.QCI**
5QI-specific downlink data rate	(**N.ThpVol.DL.DuCell5QI** - **N.ThpVol.DL.LastSlot.DuCell5QI**)/ **N.ThpTime.DL.RmvLastSlot.DuCell5QI**
5QI-specific average uplink delay at the MAC layer	**N.MAC.UL.Cell5QI.Delay** /**N.MAC.UL.Packets.Cell5QI**
5QI-specific average downlink delay at the MAC layer	**N.MAC.DL.Cell5QI.Delay** / **N.MAC.DL.Packets.Cell5QI**

6.3 Requirements

6.3.1 Licenses

None

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-specific PDCCH aggregation level optimization	**PUSCH_DTX_BASED_PDCCH_ADJ_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter, **DL_BLER_TAR_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter, or the **gNBDUSchParamGroup**.***PdcchSinrOffset*** parameter	*Service-differentiated Experience Guarantee*	QCI-specific CCE aggregation level adaptation takes effect only when any of the following conditions is met: ● The **PUSCH_DTX_BASED_PDCCH_ADJ_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter is selected. ● The **DL_BLER_TAR_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter is selected. ● The **gNBDUSchParamGroup**.***PdcchSinrOffset*** parameter is set to a value other than 0.

6.3.2.2 Mutually Exclusive Functions

None

6.3.2.3 Function Impacts

None

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. For DBS3900 LampSite base stations, the BBU must be BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.3.4 Others

None

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Table 6-3 describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 6-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	CCE_AGG_LEVEL_ADAPT_SW	Select this option if QCI-specific CCE aggregation level adaptation is required.

Parameter Name	Parameter ID	Option	Setting Notes
CCE Allocation Failure Rate Upper Threshold	**gNBDUSchParamGroup**.***CceAllocFailRateUpperThld***	None	Use the recommended value.
CCE Allocation Failure Rate Lower Threshold	**gNBDUSchParamGroup**.***CceAllocFailRateLowerThld***	None	Use the recommended value.

6.4.1.2 Using MML Commands

Before using MML commands, refer to **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

This function takes effect 5 to 10 minutes after the related parameter configurations in the **gNBDUSchParamGroup** and **NRDUCellQciBearer** MOs are modified.

Activation Command Examples

QCI-specific CCE aggregation level adaptation

```
//Turning on the CCE Aggregation Level Adaptation Switch and setting CCE Allocation Failure Rate Upper Threshold and CCE Allocation Failure Rate Lower Threshold
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=CCE_AGG_LEVEL_ADAPT_SW-1,
CceAllocFailRateUpperThld=15,CceAllocFailRateLowerThld=5;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the CCE Aggregation Level Adaptation Switch
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=CCE_AGG_LEVEL_ADAPT_SW-0;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

Observe whether the uplink or downlink CCE allocation success rate increases after (1) QCI-specific CCE aggregation level adaptation is enabled together with (2) any optimization of "QCI-specific PDCCH aggregation level optimization" (3) in scenarios with a high CCE allocation failure rate in a cell. If the uplink or downlink CCE allocation success rate increases, this function has taken effect.

Indicator Name	Definition
Uplink CCE allocation success rate	(**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)/**N.CCE.UL.AllocReq.Num**
Downlink CCE allocation success rate	(**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)/**N.CCE.DL.AllocReq.Num**

6.4.3 Network Monitoring

The indicators listed in **6.2 Network Analysis** can be used to monitor the benefits and impacts of this function.

7 MAC CE-based Rate Adjustment for Data Services

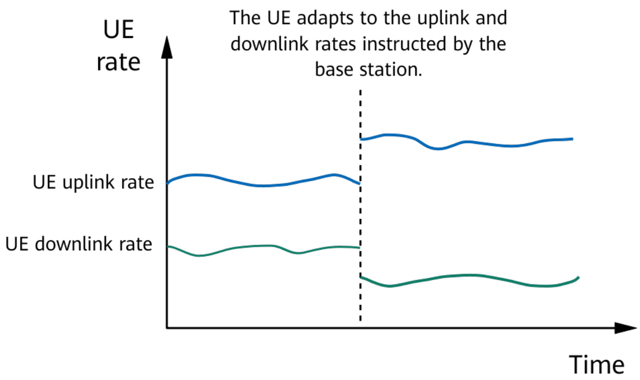
7.1 Principles

To ensure smooth user experience with live streaming or cloud gaming services, UE rates must be adapted to variant air interface conditions in real time. As such, MAC CE-based rate adjustment for data services is introduced. MAC CE stands for Media Access Control control element, which is used for control signaling at the MAC layer between the gNodeB and UEs.

The process of this function is as follows:

1. A UE sends a rate adjustment request.
2. After receiving the request, the gNodeB indicates the maximum uplink and downlink UE rates through the MAC CE based on the uplink and downlink air interface quality.
3. The UE adjusts the uplink and downlink bit rates quickly according to the base station's instruction to match the available uplink and downlink rates. See **Figure 7-1**.

Figure 7-1 Base-station-assisted UE rate adjustment



MAC CE-based rate adjustment for data services is controlled by the **ANBR_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter. It also involves the following parameter configurations:

- The transmission interval of a recommended rate can be configured by using the **gNBQciBearer**.***AnbrInterval*** parameter. The recommended parameter value is **1**, which corresponds to an effective value of 40 ms.
- The rate adjustment mode can be configured by using the **gNBQciBearer**.***AnbrMode*** parameter.
  - If this parameter is set to **UL_ONLY**, only uplink rate adjustment is supported.
  - If this parameter is set to **DL_ONLY**, only downlink rate adjustment is supported.
  - If the parameter is set to **BOTH_SUPPORT**, bidirectional rate adjustment is supported.
- The rate adjustment multiplier can be configured by using the **NRDUCellQciBearer**.***AnbrMultiplier*** parameter. The value **X40** is recommended.
- In "MAC CE-based rate adjustment for data services", the uplink and downlink resource ranges for estimation can be adjusted by flexibly configuring the **NRDUCellUlSch**.***AnbrResFactor*** and **NRDUCellDlSch**.***AnbrResFactor*** parameters. In this way, the recommended uplink and downlink rates better match the actual available uplink and downlink rates. Specifically:
  - The uplink resource range for estimation by the gNodeB is set by using the **NRDUCellUlSch**.***AnbrResFactor*** parameter.
  - The downlink resource range for estimation by the gNodeB is set by using the **NRDUCellDlSch**.***AnbrResFactor*** parameter.

  The recommended settings of the preceding parameters vary with cell bandwidths. For details, see **7.4.1.1 Data Preparation**.

This function applies to data service scenarios. For details about MAC CE-based rate adjustment for VoNR services (using bearers with 5QI 1) and ViNR services (using bearers with 5QI 2), see section "MAC CE-based Rate Adjustment" in *VoNR and ViNR*.

7.2 Network Analysis

7.2.1 Benefits

This function can improve the experience of data services such as live streaming and cloud gaming, guarantee the data service rate, and reduce frame freezing.

7.2.2 Impacts

The impacts between this function and other functions are described in **7.3.2.3 Function Impacts**.

When **gNBQciBearer**.***AnbrMode*** is set to:

- **UL_ONLY**:

- Under a light uplink network load (**Uplink Resource Block Utilizing Rate (DU)** < 20%), the average uplink throughput (**User Uplink Average Throughput (DU)**) of the related bearer increases.
- Under a heavy uplink network load (**Uplink Resource Block Utilizing Rate (DU)** > 70%), the average uplink throughput (**User Uplink Average Throughput (DU)**) of the related bearer decreases.

- **DL_ONLY**:
  - Under a light downlink network load (**Downlink Resource Block Utilizing Rate (DU)** < 20%), the average downlink throughput (**User Downlink Average Throughput (DU)**) of the related bearer increases.
  - Under a heavy downlink network load (**Downlink Resource Block Utilizing Rate (DU)** > 70%), the average downlink throughput (**User Downlink Average Throughput (DU)**) of the related bearer decreases.
- **BOTH_SUPPORT**:
  - Under a light network load in the uplink or downlink, the average throughput of the related bearer increases in the direction.
  - Under a heavy network load in the uplink or downlink, the average throughput of the related bearer decreases in the direction.

The variation of the average uplink or downlink throughput of a bearer is related to factors such as the air interface capability and service rate adjustment capability.

7.3 Requirements

7.3.1 Licenses

None

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

None

7.3.2.2 Mutually Exclusive Functions

None

7.3.2.3 Function Impacts

None

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.3.4 Others

- UEs must support the **recommendedBitRate**, **recommendedBitRateQuery**, and **recommendedBitRateMultiplier-r16** capabilities, which is true when these fields are present in *MAC-ParametersCommon* and all the fields are set to **supported**.
- The UE and service server support MAC CE-based rate adjustment.

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

Table 7-1 describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 7-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Service Differentiation Algorithm Switch	**NRDUCellQciBearer**.***ServiceDiffAlgoSwitch***	ANBR_SW	Select this option.
ANBR Interval	**gNBQciBearer**.***AnbrInterval***	None	Set this parameter to **1**.
ANBR Mode	**gNBQciBearer**.***AnbrMode***	None	Set this parameter to **UL_ONLY** to enable uplink rate adjustment. Set this parameter to **DL_ONLY** to enable downlink rate adjustment. Set this parameter to **BOTH_SUPPORT** to enable uplink rate adjustment and downlink rate adjustment.
ANBR Multiplier	**NRDUCellQciBearer**.***AnbrMultiplier***	None	Set this parameter to **X40**.
ANBR Resource Factor	**NRDUCellUlSch**.***AnbrResFactor***	None	• You are advised to set this parameter to **40** if the cell bandwidth ranges from 60 MHz to 100 MHz. • You are advised to set this parameter to **20** if the cell bandwidth is 40 MHz or 50 MHz. • You are advised to set this parameter to **10** if the cell bandwidth is 20 MHz or 30 MHz.

Parameter Name	Parameter ID	Option	Setting Notes
ANBR Resource Factor	**NRDUCellDlSch**.***AnbrResFactor***	None	• You are advised to set this parameter to **40** if the cell bandwidth ranges from 60 MHz to 100 MHz. • You are advised to set this parameter to **20** if the cell bandwidth is 40 MHz or 50 MHz. • You are advised to set this parameter to **10** if the cell bandwidth is 20 MHz or 30 MHz.

7.4.1.2 Using MML Commands

Before using MML commands, refer to **7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling MAC CE-based rate adjustment for data services
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, ServiceDiffAlgoSwitch=ANBR_SW-1;
//Configuring the MAC CE-based rate adjustment interval for data services
MOD GNBQCIBEARER: Qci=9, AnbrInterval=1;
//Configuring the MAC CE-based rate adjustment mode for data services (using uplink rate adjustment as an example)
MOD GNBQCIBEARER: Qci=9, AnbrMode=UL_ONLY;
//Configuring the multiplier for MAC CE-based rate adjustment for data services
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, AnbrMultiplier=X40;
//Setting the resource factor for uplink ANBR using the scenario where the cell bandwidth ranges from 60 MHz to 100 MHz as an example
MOD NRDUCELLULSCH: NrDuCellId=0, AnbrResFactor=40;
//Setting the resource factor for downlink ANBR using the scenario where the cell bandwidth ranges from 60 MHz to 100 MHz as an example
MOD NRDUCELLDLSCH: NrDuCellId=0, AnbrResFactor=40;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling MAC CE-based rate adjustment for data services
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, ServiceDiffAlgoSwitch=ANBR_SW-0;
```


7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

After this function is enabled, observe the values of the following counters. If the values are not 0, this function has taken effect on the base station side.

- Use the **N.MAC.DL.DlAnbr.Num.DuCell5QI** counter to monitor the number of downlink ANBR messages for services with a specific 5QI.
- Use the **N.MAC.DL.UlAnbr.Num.DuCell5QI** counter to monitor the number of uplink ANBR messages for services with a specific 5QI.

7.4.3 Network Monitoring

The indicators described in **7.2.2 Impacts** can be used to monitor the impact of this function.

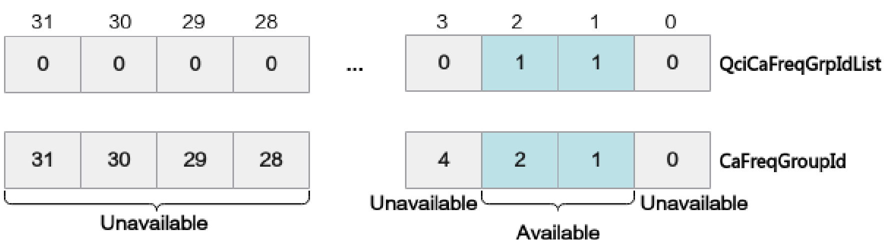
8 QoS-based CA Policy

8.1 Principles

In CA scenarios, QoS-based CA policy enables a gNodeB to configure differentiated CA frequency combinations for UEs with specific QCIs. For details about CA, see *Carrier Aggregation* and *CA Experience Improvement*.

The QoS-based CA policy function is controlled by the **QOS_CA_POLICY_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter, and the **NRCellQciBearer**.***QciCaFreqGrpIdList*** parameter is used to configure differentiated CA frequency groups for UEs with QCI-specific bearers. Bits 0 to 31 from right to left of the **NRCellQciBearer**.***QciCaFreqGrpIdList*** parameter respectively map to the availability status of the CA frequency groups with IDs (specified by the **gNBCaFrequency**.***CaFreqGroupId*** parameter) ranked in the ascending order for the local base station. For details, see **Figure 8-1**. In the cells on which UEs with QCI-specific bearers camp, the bit value of 1 indicates that the corresponding CA frequency group is available to these UEs, while the bit value of 0 indicates the opposite.

Figure 8-1 CA frequency group ID list configuration



NOTE

- If multiple QCI-specific bearers are set up for a UE and the **NRCellQciBearer**.***QciCaFreqGrpIdList*** parameter is configured for all QCIs, the UE can use the CA frequency groups of all its QCIs.
- A frequency in a CA frequency group available to a UE can be added as an SCell frequency for the UE provided that the group contains the frequency of the cell on which the UE camps.

The following is the procedure for enabling this function:

1. When CA is enabled, select the **QOS_CA_POLICY_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter on the gNodeB serving the PCell. For details about the procedure for enabling CA, see *Carrier Aggregation* and *CA Experience Improvement*.
2. The **gNBCaFrequency** MO is used to configure a limited frequency group.

   NOTE

   Limited frequency group: contains only some CA frequencies of the gNodeB.
3. Set the **NRCellQciBearer**.***QciCaFreqGrpIdList*** parameter to differentiated values for the UEs with specific QCIs in all CA cells served by the gNodeB. If a UE with QCI-specific bearers uses one of these cells as its PCell, the gNodeB evaluates the CA frequency group that can be used by the UE based on **QCI CA Frequency Group ID List** and enables differentiated CA frequency combinations to take effect.

NOTE

QoS-based CA policy is configured and processed on the gNodeB serving the PCell.

If the QCI changes after the UEs are admitted and the values of **NRCellQciBearer**.***QciCaFreqGrpIdList*** are inconsistent between the original and new QCIs, the CA frequency combinations of the UEs will change. In this case, the base station evaluates whether the new CA frequency group completely supports the configured CA frequency combinations. If not, the configured SCCs are retained according to the following priority on the basis of the new frequency group and the SCCs that are not supported are removed:

1. SCC in the CA frequency combination with more SCCs
2. SCC in the CA frequency combination with a larger total downlink bandwidth
3. SCC in the CA frequency combination with a smaller value of the **gNBCaFrequency**.***CaFreqGroupId*** parameter

8.2 Network Analysis

8.2.1 Benefits

The QoS-based CA policy function enables operators to configure differentiated CA frequency combinations for CA UEs with different QCIs, thereby providing differentiated service experience guarantee for them. As the distribution of UEs with different QCIs varies in a cell, the values of counters related to QCI-level user-perceived rates in the cell may not be significantly differentiated.

8.2.2 Impacts

The impacts between this function and other functions are described in **8.3.2.3 Function Impacts**.

After QoS-based CA policy takes effect, if CA UEs with a specific QCI use a limited frequency group, the CA frequency combinations that take effect are limited, and the uplink and downlink peak rates of these CA UEs decrease compared with those when the frequency group is not limited.

8.3 Requirements

8.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-091230	QoS-based CA Policy	NR0S00QSCP10	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter **INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	QoS-based CA policy depends on CA. At least one of the following functions must be enabled: ● Intra-band CA ● Intra-FR inter-band CA ● Inter-gNodeB CA ● Inter-FR CA

Function Name	Function Switch	Reference	Description
Intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter **INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	
Inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*CA Experience Improvement*	
Inter-FR CA (trial)	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	QoS-based CA policy depends on CA. At least one of the following functions must be enabled: ● Intra-band CA ● Intra-FR inter-band CA ● Inter-gNodeB CA ● Inter-FR CA

8.3.2.2 Mutually Exclusive Functions

None

8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NSA carrier combination selection based on user experience	• On the NR side: **NSA_LTE_SEL_OPT_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter selected • On the LTE side: **NSA_LTE_FDD_SEL_OPT_SW** or **NSA_LTE_TDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter selected	*EPC-based NSA Experience Improvement*	When both NSA carrier combination selection based on user experience and QoS-based CA policy are enabled, if the LTE-side parameter **NsaDcAlgoParam**.***NsaDcNrCaBwEvalInd*** is set to **NR_EVAL_ALLOWED** or **NR_EVAL_NOT_ALLOWED_IN_SCG_ADD**, CA may not take effect for UEs with a specific QCI and using the limited frequency group because the LTE side does not transfer QCIs to the NR side. In this case, the **NSA_L2N_QCI_TRANSFER_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter can be selected on the LTE side. The LTE side will transfer the UE QCI list to the NR side when requesting experience evaluation assistance from the NR side, thereby avoiding affecting the QoS-based CA policy function.

8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. For DBS3900 LampSite base stations, the BBU must be BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

8.3.4 Others

- UEs

  UEs comply with 3GPP Release 15 or later and support CA.
- Core network

  The core network complies with 3GPP Release 15 or later and supports QCI configuration.

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

Table 8-1 describes the parameters used for function activation. This table does not describe parameters related to cell establishment. For details about data preparation for CA functions, see *Carrier Aggregation* and *CA Experience Improvement*.

Table 8-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Networking Option Opt Sw	**gNodeBParam**.***NetworkingOptionOptSw***	QOS_CA_POLICY_SW	Select this option if the QoS-based CA policy function needs to be enabled.

Parameter Name	Parameter ID	Option	Setting Notes
CA Frequency Group ID	**gNBCaFrequency**.***CaFreqGroupId***	N/A	Set this parameter based on the network plan. It is recommended that all frequencies planned for CA for a base station be configured in the same frequency group.
Downlink NARFCN	**gNBCaFrequency**.***DlNarfcn***	N/A	Set this parameter based on the network plan.
SSB Frequency Position	**gNBCaFrequency**.***SsbFreqPos***	N/A	Set this parameter based on the network plan.
QoS Class Identifier	**NRCellQciBearer**.***Qci***	None	Set this parameter based on the network plan.
QCI CA Frequency Group ID List	**NRCellQciBearer**.***QciCaFreqGrpIdList***	None	Set this parameter based on the network plan.
Multi Networking Option Opt Switch	**EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw***	NSA_L2N_QCI_TRANSFER_SW	Select this option on the LTE side when the LTE-side parameter **NsaDcAlgoParam**.***NsaDcNrCaBwEvalInd*** is set to **NR_EVAL_ALLOWED** or **NR_EVAL_NOT_ALLOWED_IN_SCG_ADD** in scenarios where both NSA carrier combination selection based on user experience and QoS-based CA policy are enabled.

8.4.1.2 Using MML Commands

Before using MML commands, refer to **8.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- For details about MML command examples for CA functions, see *Carrier Aggregation* and *CA Experience Improvement*.
- QoS-based CA policy

```
//Turning on the switch for QoS-based CA policy
MOD GNODEBPARAM: NetworkingOptionOptSw=QOS_CA_POLICY_SW-1;

//Adding CA frequency group configurations in tri-band networking of n78, n41, and n1 as an example (The downlink frequency and SSB frequency-domain position must be the same as those in the NRDUCell MO and the following configurations are for example purposes only.)
//Configuring CA frequency group 0
ADD GNBCAFREQUENCY: CaFreqGroupId=0, DlNarfcn=627264, SsbFreqPos=7881;
ADD GNBCAFREQUENCY: CaFreqGroupId=0, DlNarfcn=504990, SsbFreqPos=7812;
ADD GNBCAFREQUENCY: CaFreqGroupId=0, DlNarfcn=152650, SsbFreqPos=3812;

//Configuring CA frequency group 1
ADD GNBCAFREQUENCY: CaFreqGroupId=1, DlNarfcn=504990, SsbFreqPos=7812;
ADD GNBCAFREQUENCY: CaFreqGroupId=1, DlNarfcn=152650, SsbFreqPos=3812;

//Configuring CA frequency group 2
ADD GNBCAFREQUENCY: CaFreqGroupId=2, DlNarfcn=627264, SsbFreqPos=7881;
ADD GNBCAFREQUENCY: CaFreqGroupId=2, DlNarfcn=152650, SsbFreqPos=3812;

//Configuring QCI CA Frequency Group ID List in different cells (Assume that cells 0, 1, and 2 work in bands n78, n41, and n1, respectively, a limited frequency group is used for UEs with QCI 3, and the frequency group used by UEs with QCI 2 is not limited.)
MOD NRCELLQCIBEARER: NrCellId=0, Qci=3, QciCaFreqGrpIdList=4;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=2, QciCaFreqGrpIdList=4294967295;
MOD NRCELLQCIBEARER: NrCellId=1, Qci=3, QciCaFreqGrpIdList=2;
MOD NRCELLQCIBEARER: NrCellId=1, Qci=2, QciCaFreqGrpIdList=4294967295;
MOD NRCELLQCIBEARER: NrCellId=2, Qci=3, QciCaFreqGrpIdList=4;
MOD NRCELLQCIBEARER: NrCellId=2, Qci=2, QciCaFreqGrpIdList=4294967295;

//(Optional) Turning on the switch for "LTE-to-NR QCI transfer in NSA networking" on the LTE side when NSA carrier combination selection based on user experience is enabled in NSA networking
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=NSA_L2N_QCI_TRANSFER_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for QoS-based CA policy
MOD GNODEBPARAM: NetworkingOptionOptSw=QOS_CA_POLICY_SW-0;
```

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

In NSA networking, observe the message tracing result as follows:

Step 1 Log in to the MAE-Access and choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type** > **LTE** > **Application Layer** > **X2 Interface Trace**.

Step 3 Enable a UE to access a cell. After a bearer with a specific QCI or 5QI is set up, enable the UE to perform services.

Step 4 Observe the SGNB_ADD_REQ_ACK message delivered when the base station adds an SCell. If the **frequencyBandList** field in the *SCellConfig* IE only includes frequencies of a limited frequency group, this function has taken effect.

----End

In SA networking, observe the message tracing result as follows:

Step 1 Log in to the MAE-Access and choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type** > **NR** > **Application Layer** > **Uu Interface Trace**.

Step 3 Enable a UE to access a cell. After a bearer with a specific QCI or 5QI is set up, enable the UE to perform services.

Step 4 Observe the RRCReconfiguration message delivered when the base station adds an SCell. If the **frequencyBandList** field in the *SCellConfig* IE only includes frequencies of a limited frequency group, this function has taken effect.

----End

8.4.3 Network Monitoring

The network monitoring is consistent between this function and CA. For details, see the network monitoring of functions in *Carrier Aggregation* and *CA Experience Improvement*.

9 Flexible Experience Differentiation

9.1 Principles

To enable differentiated and customizable peak user-perceived rates, flexible experience differentiation is introduced. This function is controlled by the **FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter. This function includes the following sub-functions:

- Data rate limitation in flexible experience differentiation: This sub-function allows differentiated data rate proportions to be configured for users with different privilege levels, thereby achieving differentiated and customizable peak user-perceived rates. For details, see **9.1.1 Data Rate Limitation in Flexible Experience Differentiation**.
- Data rate reporting in flexible experience differentiation: This sub-function allows the uplink or downlink data rate to be periodically reported to the core network. For details, see **9.1.2 Data Rate Reporting in Flexible Experience Differentiation**.

Flexible experience differentiation takes effect only for SA UEs.

9.1.1 Data Rate Limitation in Flexible Experience Differentiation

"Data rate limitation in flexible experience differentiation" enables differentiated and customizable data rates by (1) setting different privilege levels for users with different QCIs, (2) configuring different target data rate proportions and minimum guaranteed bandwidths for privilege levels, and (3) periodically calculating the uplink or downlink peak user rate. The specific procedure is as follows:

1. Based on UE and network capabilities, the associated carrier combination is configured in compliance with the optimal carrier selection procedure.
2. The uplink or downlink peak rates of UEs at their current locations are estimated based on carrier measurement information, related configurations, and the factor for estimating differentiated uplink or downlink target rates.

3. Based on minimum guaranteed bandwidths, the UE and network capabilities are mapped to user privilege levels, and the uplink or downlink peak data rate of each privilege level is periodically calculated.
4. Data rate control is periodically performed based on the uplink or downlink peak data rates of the privilege levels corresponding to the QCIs of UEs. If the rate threshold for exiting flexible experience differentiation in the uplink or downlink is not 0, the base station makes the following comparisons before performing rate control on UEs:
   - Compares the uplink peak rates of UEs at their current locations with the rate threshold for exiting flexible experience differentiation in the uplink.
     - If the estimated uplink peak rates of UEs are greater than or equal to the rate threshold for exiting flexible experience differentiation in the uplink, the base station performs uplink rate control on the UEs based on the uplink peak rates of the privilege levels corresponding to the QCIs of UEs.
     - If the estimated uplink peak rates of UEs are less than the rate threshold for exiting flexible experience differentiation in the uplink, the base station does not perform uplink rate control on the UEs within this period.
   - Compares the downlink peak rates of UEs at their current locations with the rate threshold for exiting flexible experience differentiation in the downlink.
     - If the estimated downlink peak rates of UEs are greater than or equal to the rate threshold for exiting flexible experience differentiation in the downlink, the base station performs downlink rate control on the UEs based on the downlink peak rates of the privilege levels corresponding to the QCIs of UEs.
     - If the estimated downlink peak rates of UEs are less than the rate threshold for exiting flexible experience differentiation in the downlink, the base station does not perform downlink rate control on the UEs within this period.

Specifically:

- The **NRDUCellQciBearer**.***EquityLevel*** parameter specifies the privilege level of each QCI-specific bearer.
- The **NRDUCellUuPrfmOpt**.***DiffTargetRateCalcPeriod*** parameter specifies the calculation period and is defaulted to 2s.
- The **NRDUCellPusch**.***UlDiffTargetRateEstFactor*** parameter specifies the factor for estimating the differentiated uplink target rate. The recommended value is **90**.
- The **NRDUCellUuPrfmOpt**.***DiffTargetRateEstFactor*** parameter specifies the factor for estimating the differentiated downlink target rate. The recommended value is **90**.
- The **NRDUCellPusch**.***UlFlexExpDiffExitRateThld*** parameter specifies the rate threshold for exiting flexible experience differentiation in the uplink.
- The **NRDUCellRsvdExt04**.***RsvdParam56*** parameter specifies the rate threshold for exiting flexible experience differentiation in the downlink.

- For the uplink, the **UL_UNLIMITED_RATE_SW** option of the **gNBFlexExpDiffCfg**.***EquityLevelAlgoSw*** parameter specifies whether "data rate limitation in flexible experience differentiation" takes effect for users with a privilege level. This option is deselected by default. If this option is deselected, "data rate limitation in flexible experience differentiation" takes effect for users with the corresponding privilege level. This option can be selected if this function is not required for users with the corresponding privilege level.
- For the downlink, the **DL_UNLIMITED_RATE_SW** option of the **gNBFlexExpDiffCfg**.***EquityLevelAlgoSw*** parameter specifies whether "data rate limitation in flexible experience differentiation" takes effect for users with a privilege level. This option is deselected by default. If this option is deselected, "data rate limitation in flexible experience differentiation" takes effect for users with the corresponding privilege level. This option can be selected if this function is not required for users with the corresponding privilege level.
- **Table 9-1** describes the flexible experience differentiation configurations for each privilege level.

Table 9-1 Flexible experience differentiation configurations

Parameter Name	Parameter ID	Description	Impact
Equity Level	**NRDUCellQciBearer**.***EquityLevel***	If this parameter is set to a value ranging from 0 to 5, a smaller value of this parameter indicates a higher privilege level.	More privilege levels lead to more user-perceived data rate levels.
		If this parameter is set to 255, flexible experience differentiation does not take effect for the QCI.	
Equity Level	**gNBFlexExpDiffCfg**.***EquityLevel***	Set this parameter to a value ranging from 0 to 5. A smaller value of this parameter indicates a higher privilege level.	

Parameter Name	Parameter ID	Description	Impact
UL Target Rate Ratio	**gNBFlexExpDiffCfg**.***UlTargetRateRatio***	Set this parameter to a value ranging from 1 to 100 in percentage. The value of this parameter is the result of the uplink available peak data rate of a QCI-specific user divided by the current network's maximum capability.	In conditions of the same network and UE capabilities, smaller data rate proportions of a privilege level result in: (1) more significant differences in the uplink or downlink peak data rate between this privilege level and higher privilege levels; (2) lower uplink or downlink peak data rate of users at this privilege level, potentially impacting user experience.
DL Target Rate Ratio	**gNBFlexExpDiffCfg**.***DlTargetRateRatio***	Set this parameter to a value ranging from 1 to 100 in percentage. The value of this parameter is the result of the downlink available peak data rate of a QCI-specific UE divided by the current network's maximum capability.	

Parameter Name	Parameter ID	Description	Impact
UL Min Guarantee Bandwidth	**gNBFlexExpDiffCfg**.***UlMinGteeBw***	Set this parameter to a value ranging from 0 to 1000, which indicates the minimum uplink guaranteed bandwidth for the associated privilege level. If this parameter is set to 0, the UE and network capabilities are not verified.	Given the same network and UE capabilities, a larger minimum guaranteed bandwidth may result in less significant differences in the uplink or downlink peak data rate between privilege levels. ● If the bandwidth configured for a user cannot reach the minimum guaranteed bandwidth of any level, the uplink or downlink peak data rate of each privilege level is the current maximum available data rate for the user, with no difference between privilege levels. ● If this parameter is set to 0, UE and network capabilities are not verified. The uplink or downlink peak user-perceived rate proportions of different privilege levels are basically consistent with the configured target uplink or downlink data rate proportions. ● If the minimum guaranteed bandwidths of different privilege levels are not configured based on the target uplink or downlink data rate proportions, the uplink or downlink peak user-perceived rate proportions of
DL Min Guarantee Bandwidth	**gNBFlexExpDiffCfg**.***DlMinGteeBw***	Set this parameter to a value ranging from 0 to 1000, which indicates the minimum downlink guaranteed bandwidth for the associated privilege level. If this parameter is set to 0, the UE and network capabilities are not verified.	

Parameter Name	Parameter ID	Description	Impact
			different privilege levels may not be fully consistent with the configured target uplink or downlink data rate proportions.

NOTE

- Data rate limitation in flexible experience differentiation takes effect only when the **FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter is selected.
- When multiple bearers are set up for a UE, the function parameter settings corresponding to the UE's QCI with the highest priority take effect. For details about QCI configurations of UEs and QCI priorities, see *QoS Management*.
- If no privilege level is configured for any QCI-specific bearer of a UE, the base station does not estimate the data rate or control the peak data rate in real time for the UE.
- If the **NRDUCellQciBearer**.***EquityLevel*** parameter is set to a value other than **255**, the associated **gNBFlexExpDiffCfg**.***EquityLevel*** parameter must already be configured.
- If multiple **gNBFlexExpDiffCfg** MOs are configured, any two of them must meet the following requirements: For the MO with a larger value of the **gNBFlexExpDiffCfg**.***EquityLevel*** parameter, the values of its associated **gNBFlexExpDiffCfg**.***UlTargetRateRatio***, **gNBFlexExpDiffCfg**.***DlTargetRateRatio***, **gNBFlexExpDiffCfg**.***UlMinGteeBw***, and **gNBFlexExpDiffCfg**.***DlMinGteeBw*** parameters must be less than or equal to those in the other MO.

9.1.2 Data Rate Reporting in Flexible Experience Differentiation

With data rate reporting in flexible experience differentiation, the base station reports the estimated uplink or downlink data rate for each privilege level to the core network based on the negotiation result between the base station and the core network. This function takes effect as follows:

1. The core network sends the base station an indication specifying whether the QoS flow supports reporting of the maximum available data rate IE. For each established, added, or modified QoS flow, if the PDU SESSION RESOURCE SETUP REQUEST or PDU SESSION RESOURCE MODIFY REQUEST message contains a request for monitoring the maximum available data rate IE, the base station stores this monitoring request and monitors the available data rate.
2. After receiving the QoS-flow-level indication, the base station performs multi-flow combination and processing, and then determines whether to report an estimated uplink or downlink data rate based on the corresponding indication.
3. If reporting is required, the base station reports the periodically maintained data rate information for each privilege level to the core network through a private GTP-U extension header in the QoS flow.

Data rate reporting in flexible experience differentiation is controlled by the **FLEX_EXP_DIFF_RPT_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter.

NOTE

The **FLEX_EXP_DIFF_RPT_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter can be selected only when the **FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter is selected.

9.2 Network Analysis

9.2.1 Benefits

After this function is enabled, differentiated peak user-perceived rates can be customized for users at different privilege levels on the network.

- Given the same network conditions, the uplink or downlink peak user-perceived rates for different privilege levels of a single user match the associated flexible experience differentiation configurations. The details are as follows:
  - For each privilege level of a user, if the uplink or downlink minimum guaranteed bandwidth is set to 0, the uplink or downlink peak user-perceived rate meets the configured target uplink or downlink data rate proportion.
  - For a privilege level of a user, if the uplink or downlink minimum guaranteed bandwidth is not set to 0, the uplink or downlink peak user-perceived rate is affected by (1) the target uplink or downlink data rate proportion and (2) the uplink or downlink minimum guaranteed bandwidth. In this case, the uplink or downlink peak user-perceived rate of each privilege level of the user may not match the configured target uplink or downlink data rate proportion.
- When users with different privilege levels perform full-buffer services simultaneously under the same network conditions, the users with different privilege levels have differentiated uplink or downlink peak user-perceived rates which cannot precisely match the associated proportions in flexible experience differentiation configurations. This is caused by multiple factors such as flexible experience differentiation configuration, user scheduling priority, and data rate estimation.

NOTE

The same network conditions indicate that the UE locations, cells that UEs access, spectral efficiency, network load, UE capabilities, and other factors are the same.

9.2.2 Impacts

The impacts between this function and other functions are described in **9.3.2.3 Function Impacts**.

After data rate limitation in flexible experience differentiation is enabled:

- In the following scenarios, the limited uplink or downlink peak data rate of some UEs prolongs the user scheduling duration in conditions of the same data volume. As a result, the uplink or downlink RB usage in a cell, average uplink or downlink UE throughput, average uplink or downlink cell throughput, uplink or downlink traffic volume in a cell, and other indicators may decrease. In addition, the average downlink packet delay at the MAC layer, average downlink packet delay at the RLC layer, and other indicators may increase. For the definitions of related indicators, see **Table 9-2**. A higher penetration rate of UEs without rate limitation leads to a smaller impact of this function on KPIs.

– Due to rapid changes in factors such as network loads and channel conditions, the uplink or downlink peak data rate of each privilege level, which is calculated periodically during the execution of this function, becomes inaccurate.

– The configured target uplink and downlink data rate proportions are excessively low. In addition, the uplink and downlink minimum guaranteed bandwidths are lower than the bandwidths available for the UE before this function is enabled.

Table 9-2 Indicator definitions related to the impact analysis of data rate limitation in flexible experience differentiation

Indicator Name	Definition
Average downlink UE throughput	**User Downlink Average Throughput (DU)**
Average uplink UE throughput	**User Uplink Average Throughput (DU)**
Average downlink cell throughput	**Cell Downlink Average Throughput (DU)**
Average uplink cell throughput	**Cell Uplink Average Throughput (DU)**
Downlink RB usage	**Downlink Resource Block Utilizing Rate (DU)**
Uplink RB usage	**Uplink Resource Block Utilizing Rate (DU)**
Downlink traffic volume in a cell	**Downlink Traffic Volume (DU)**
Uplink traffic volume in a cell	**Uplink Traffic Volume (DU)**
Average downlink packet delay at the MAC layer	(**N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num**)/1000
Average downlink packet delay at the RLC layer	(**N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num**)/1000

- Parameters related to flexible experience differentiation have the following impacts on the network.

Parameter Name	Parameter ID	Impact on the Network
Diff Target Rate Calculation Period	**NRDUCellUuPrfmOpt**.***DiffTargetRateCalcPeriod***	• A smaller value of this parameter improves the accuracy of the calculated differentiated target data rate but increases the computational overhead. • A larger value of this parameter leads to slower updates of the differentiated target data rate and increases the error of the calculated target data rate.
UL Target Rate Ratio	**gNBFlexExpDiffCfg**.***UlTargetRateRatio***	• A larger value of this parameter leads to a higher uplink peak perceived rate of users at the associated privilege level. • A smaller value of this parameter leads to a lower uplink peak perceived rate of users at the associated privilege level.
DL Target Rate Ratio	**gNBFlexExpDiffCfg**.***DlTargetRateRatio***	• A larger value of this parameter leads to a higher downlink peak perceived rate of users at the associated privilege level. • A smaller value of this parameter leads to a lower downlink peak perceived rate of users at the associated privilege level.
UL Min Guarantee Bandwidth	**gNBFlexExpDiffCfg**.***UlMinGteeBw***	• A larger value of this parameter leads to a higher uplink peak perceived rate of users at the associated privilege level. • A smaller value of this parameter leads to a lower uplink peak perceived rate of users at the associated privilege level.
DL Min Guarantee Bandwidth	**gNBFlexExpDiffCfg**.***DlMinGteeBw***	• A larger value of this parameter leads to a higher downlink peak perceived rate of users at the associated privilege level. • A smaller value of this parameter leads to a lower downlink peak perceived rate of users at the associated privilege level.

Data rate reporting in flexible experience differentiation: None

9.3 Requirements

9.3.1 Licenses

In TDD, the license requirements for flexible experience differentiation are as follows:

Feature ID	Feature Name	Model	Sales Unit
FOFD-100210	Flexible Rate Grading (trial)	NR0SFLEXGR10	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.3.2.1 Prerequisite Functions

None

9.3.2.2 Mutually Exclusive Functions

None

9.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink UE-AMBR-based rate limitation for non-GBR services	**UL_AMBR_RATE_CTRL_SW** option of the **NRDUCellAlgoSwitch**.***ServiceDiffSwitch*** parameter	*QoS Management*	After both "uplink UE-AMBR-based rate limitation for non-GBR services" and "flexible experience differentiation" are enabled, the UE's uplink peak data rate takes the value of the target uplink data rate calculated in "flexible experience differentiation" if the uplink UE-AMBR used for "uplink UE-AMBR-based rate limitation for non-GBR services" is greater than the aforementioned calculated target uplink data rate.
Low-frequency TDD	Downlink UE-AMBR-based rate limitation for non-GBR services	**DL_AMBR_RATE_CTRL_SW** option of the **NRDUCellAlgoSwitch**.***ServiceDiffSwitch*** parameter	*QoS Management*	After both "downlink UE-AMBR-based rate limitation for non-GBR services" and "flexible experience differentiation" are enabled, the UE's downlink peak data rate takes the value of the target downlink data rate calculated in "flexible experience differentiation" if the downlink UE-AMBR used for "downlink UE-AMBR-based rate limitation for non-GBR services" is greater than the aforementioned calculated target downlink data rate.
Low-frequency TDD	Maximum rate limitation for GBR services	None	*QoS Management*	When a UE performs GBR services and "flexible experience differentiation" is enabled, the UE's uplink or downlink peak data rate takes the value of the target uplink or downlink data rate calculated in "flexible experience differentiation" if the UE's MBR or MFBR is greater than the aforementioned calculated target uplink or downlink data rate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Minimum uplink rate guarantee for non-GBR services	**gNBDUMacParamGroup.*UlGuaranteedRate***	*QoS Management*	After "flexible experience differentiation" is enabled, a UE's uplink peak data rate takes the value of the **gNBDUMacParamGroup.*UlGuaranteedRate*** parameter if this parameter is set to a non-zero value which exceeds the target uplink data rate calculated in "flexible experience differentiation". As a result, the UE's uplink data rate may not be consistent with the flexible experience differentiation configuration, which may lead to an actual data rate lower than the minimum guaranteed rate.
Low-frequency TDD	Minimum downlink rate guarantee for non-GBR services	**gNBDUMacParamGroup.*DlGuaranteedRate***	*QoS Management*	After "flexible experience differentiation" is enabled, a UE's downlink peak data rate takes the value of the **gNBDUMacParamGroup.*DlGuaranteedRate*** parameter if this parameter is set to a non-zero value which exceeds the target downlink data rate calculated in "flexible experience differentiation". As a result, the UE's downlink data rate may not be consistent with the flexible experience differentiation configuration, which may lead to an actual data rate lower than the minimum guaranteed rate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Minimum rate guarantee for GBR services	None	*QoS Management*	When a UE performs GBR services and "flexible experience differentiation" is enabled, the UE's uplink or downlink peak data rate takes the value of the GBR or GFBR configured on the core network if the UE's GBR or GFBR is greater than the target uplink or downlink data rate calculated in "flexible experience differentiation". As a result, the UE's uplink or downlink data rate may not be consistent with the flexible experience differentiation configuration, which may lead to an actual data rate lower than the minimum guaranteed rate.
Low-frequency TDD	Uplink experience-based scheduling	**UL_EXP_BASED_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	After both "uplink experience-based scheduling" and "flexible experience differentiation" take effect for a UE, the UE's uplink peak data rate takes the value of the target uplink data rate calculated in "flexible experience differentiation" if the value of the **gNBDUMacParamGroup**.***UlExpBasedSchGuarUserRate*** or **gNBDURfspConfig**.***UlExpBasedSchGuarUserRate*** parameter is greater than the aforementioned calculated target uplink data rate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink experience-based scheduling	**DL_EXP_BASED_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	After both "downlink experience-based scheduling" and "flexible experience differentiation" take effect for a UE, the UE's downlink peak data rate takes the value of the target downlink data rate calculated in "flexible experience differentiation" if the value of the **gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate*** or **gNBDURfspConfig**.***DlExpBasedSchGuarUserRate*** parameter is greater than the aforementioned calculated target downlink data rate.
Low-frequency TDD	Downlink experience satisfaction guarantee	**DL_EXP_SATIS_GUAR_SW** option of the **NRDUCellFwaQos**.***SchAlgoSwitch*** parameter	*FWA*	Assume that both downlink experience satisfaction guarantee and flexible experience differentiation are in effect for a UE. If the value of the **gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate*** or **gNBDURfspConfig**.***DlExpBasedSchGuarUserRate*** parameter exceeds the downlink target rate determined by the flexible experience differentiation function, the UE's downlink peak rate will be capped at this downlink target rate determined by the flexible experience differentiation function.
Low-frequency TDD	Service Type	**gNBQciBearer**.***ServiceType*** set to a value other than **NORMAL**	None	If the service type of a QCI-specific bearer is not common service and there are certain requirements on the uplink or downlink data rate, it is not recommended that data rate limitation in flexible experience differentiation be enabled for the QCI.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	Flexible experience differentiation does not take effect for RedCap UEs.
Low-frequency TDD	User PRB control for FWA	**PRB_CTRL_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	After both "user PRB control for FWA" and "flexible experience differentiation" are enabled, the base station estimates the uplink or downlink peak data rate based on the available resource range of FWA UEs and performs further data rate control. As a result, the difference between the estimated data rate and the actually measured data rate may increase.
Low-frequency TDD	NR DU cell resource management between network slices	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	After both "NR DU cell resource management between network slices" and "flexible experience differentiation" are enabled, the base station estimates the uplink or downlink peak data rate based on the available resource range of the slice to which the UE belongs and performs further data rate control. As a result, the difference between the estimated data rate and the actually measured data rate may increase.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RFSP-group-based RB resource control	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellCommonSw**.***RfspAlgoSwitch*** parameter	*Resource Control in NSA Networking*	After both "RFSP-group-based RB resource control" and "flexible experience differentiation" are enabled, the base station estimates the uplink or downlink peak data rate based on the available resource range of the RFSP resource group to which the UE belongs and performs further data rate control. As a result, the difference between the estimated data rate and the actually measured data rate may increase.
Low-frequency TDD	RAN sharing with common carrier	**gNBSharingMode**.***gNBMultiOpSharingMode*** set to **SHARED_FREQ**	*Multi-Operator Sharing*	After "flexible experience differentiation" is enabled in RAN sharing with common carrier scenarios, the base station estimates the uplink or downlink peak data rate based on the available resource range of the UE's operator and performs further data rate control. As a result, the difference between the estimated data rate and the actually measured data rate may increase.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter **INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	After both "CA" and "flexible experience differentiation" are enabled, the base station estimates the uplink or downlink peak data rate of each carrier of a UE. After both "CA" and "flexible experience differentiation" are enabled, the following rules also apply: (1) If the SCell activation threshold is set to a small value, the estimated uplink or downlink peak UE data rate is more accurate, but the UE power consumption increases. (2) If an SCell cannot be activated because the SCell activation threshold is set to a large value, the estimated uplink or downlink peak UE data rate may be inaccurate. It is recommended that the **NRDUCellCarrMgmt**.***CaDlScellActThldSlotNum*** and **NRDUCellCarrMgmt**.***CaUlScellActThldSlotNum*** parameters be set to **0** in this scenario.
Low-frequency TDD	Intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter **INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	
Low-frequency TDD	Inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*CA Experience Improvement*	
Low-frequency TDD	Inter-FR CA (trial)	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	Flexible experience differentiation does not take effect for inter-FR CA UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Super uplink	**SUPER_UL_TDM_SCH_SW**, **FLEX_SUPER_UPLINK_SW**, and **INTER_GNB_SUPER_UPLINK_SW** options of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	After both "super uplink" and "flexible experience differentiation" are enabled, the base station estimates the uplink peak data rate of each carrier of a UE. If a UE is configured with an SUL carrier which is not activated, the UE's estimated uplink peak data rate may be inaccurate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink non-consecutive scheduling	**UL_NON_CON_SCH_SW** option of the **NRDUCellAlgoSwitch**.***UlInconsecutiveSchSwitch*** parameter	*Uplink Scheduling*	When the **UL_NON_CON_SCH_SW** option of the **NRDUCellAlgoSwitch**.***UlInconsecutiveSchSwitch*** parameter is selected, and the **UL_UNLIMITED_RATE_SW** option of the **gNBFlexExpDiffCfg**.***EquityLevelAlgoSw*** parameter is selected for users with the associated privilege level, the estimated maximum uplink target rate may be lower than expected for uplink users with uplink non-consecutive scheduling in effect. If this occurs, the uplink rate differentiation multiple between users with different privilege levels is increased. Therefore, it is recommended that the **UL_UNLIMITED_RATE_SW** option of the **gNBFlexExpDiffCfg**.***EquityLevelAlgoSw*** parameter be deselected in scenarios with flexible experience differentiation enabled when (1) the **UL_NON_CON_SCH_SW** option of the **NRDUCellAlgoSwitch**.***UlInconsecutiveSchSwitch*** parameter is selected and (2) the differentiation multiple for uplink users is expected to match the configured value.

9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. For DBS3900 LampSite base stations, the BBU must be BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

9.3.4 Others

- Data rate limitation in flexible experience differentiation: None
- Data rate reporting in flexible experience differentiation: The core network must support the delivery of the available data rate query indication.

9.4 Operation and Maintenance

9.4.1 Data Configuration

9.4.1.1 Data Preparation

Table 9-3 describes the parameters used for function activation. This table does not describe parameters related to cell establishment. For details about data preparation for CA functions, see *Carrier Aggregation* and *CA Experience Improvement*.

Table 9-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
QCI-based Diff Service Experience Switch	**NRDUCellFeatureSw**. ***QciBasedDiffSer-vExpSw***	FLEX_EXP_DIFF_SW	Select this option if flexible experience differentiation needs to be enabled.

Parameter Name	Parameter ID	Option	Setting Notes
QCI-based Diff Service Experience Switch	**NRDUCellFeatureSw**.***QciBasedDiffServExpSw***	FLEX_EXP_DIFF_RPT_SW	It is recommended that this option be selected when the core network supports the delivery of the available data rate query indication.
Diff Target Rate Calculation Period	**NRDUCellUuPrfmOpt**.***DiffTargetRateCalcPeriod***	None	Set this parameter based on the network plan. To calculate the differentiated target data rate in a timely manner, decrease the value of this parameter.
Equity Level	**gNBFlexExpDiffCfg**.***EquityLevel***	None	Set this parameter based on the network plan.
UL Diff Target Rate Estimation Factor	**NRDUCellPusch**.***UlDiffTargetRateEstFactor***	None	Set this parameter based on the network plan.
Diff Target Rate Estimation Factor	**NRDUCellUuPrfmOpt**.***DiffTargetRateEstFactor***	None	Set this parameter based on the network plan.
UL Target Rate Ratio	**gNBFlexExpDiffCfg**.***UlTargetRateRatio***	None	Set this parameter based on the network plan.
DL Target Rate Ratio	**gNBFlexExpDiffCfg**.***DlTargetRateRatio***	None	Set this parameter based on the network plan.
UL Min Guarantee Bandwidth	**gNBFlexExpDiffCfg**.***UlMinGteeBw***	None	Set this parameter based on the network plan. If this parameter is set to **0**, the UE and network capabilities are not verified, and the target rate of each level is calculated only based on the target rate proportion.

Parameter Name	Parameter ID	Option	Setting Notes
DL Min Guarantee Bandwidth	**gNBFlexExpDiffCfg**.***DlMinGteeBw***	None	Set this parameter based on the network plan. If this parameter is set to **0**, the UE and network capabilities are not verified, and the target rate of each level is calculated only based on the target rate proportion.
Reserved Parameter 56	**NRDUCellRsvdExt04**.***RsvdParam56***	None	Set this parameter based on the network plan.
UL Flex Exp Diff Exit Rate Threshold	**NRDUCellPusch**.***UlFlexExpDiffExitRateThld***	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Equity Level Algorithms Sw	**gNBFlexExpDiffCfg**.***EquityLevelAlgoSw***	UL_UNLIMITED_RATE_SW	It is recommended that this option be deselected when the **UL_NON_CON_SCH_SW** option of the **NRDUCellAlgoSwitch**.***UlInconsecutiveSchSwitch*** parameter is selected and the differentiation multiple for uplink users is expected to match the configured value. For details, see the related descriptions in **9.3.2.3 Function Impacts**. In other scenarios, set this option based on the network plan. For the uplink, if this option is deselected and flexible experience differentiation is enabled, data rate limitation in flexible experience differentiation takes effect for users at the associated privilege level based on the associated flexible experience differentiation configuration; if this option is selected, uplink experience of users at the associated privilege level is not affected by data rate limitation.

Parameter Name	Parameter ID	Option	Setting Notes
Equity Level Algorithms Sw	**gNBFlexExpDiffCfg**.***EquityLevelAlgoSw***	DL_UNLIMITED_RATE_SW	Set this parameter based on the network plan. For the downlink, if this option is deselected and flexible experience differentiation is enabled, data rate limitation in flexible experience differentiation takes effect for users at the associated privilege level based on the associated flexible experience differentiation configuration; if this option is selected, downlink experience of users at the associated privilege level is not affected by data rate limitation.
Equity Level	**NRDUCellQciBearer**.***EquityLevel***	None	Set this parameter based on the network plan. A value other than **255** is recommended if flexible experience differentiation needs to take effect for UEs with the associated QCI. If this parameter is set to a value other than **255**, the associated **gNBFlexExpDiffCfg**.***EquityLevel*** instance must exist.
CA Activate and BWP Switch DL Thld Slot Num	**NRDUCellCarrMgmt**.***CaDlScellActThldSlotNum***	None	In CA scenarios, it is recommended that these parameters be set to **0** to improve the accuracy of data rate estimation. For details, see **9.3.2.3 Function Impacts**.
CA Deactivate and BWP Switch DL Thld Slot Num	**NRDUCellCarrMgmt**.***CaDlScellDeactThldSlotNum***	None	

Parameter Name	Parameter ID	Option	Setting Notes
CA Activate and BWP Switch UL Thld Slot Num	**NRDUCellCarrMgmt**.*CaUlScellActThld-SlotNum*	None	
CA Deactivate and BWP Switch UL Thld Slot Num	**NRDUCellCarrMgmt**.*CaUlScellDeactThld-SlotNum*	None	

9.4.1.2 Using MML Commands

Before using MML commands, refer to **9.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the flexible experience differentiation switch
MOD NRDUCELLFEATURESW: NrDuCellId=0, QciBasedDiffServExpSw=FLEX_EXP_DIFF_SW-1;
//Turning on the switch of data rate reporting in flexible experience differentiation
MOD NRDUCELLFEATURESW: NrDuCellId=0, QciBasedDiffServExpSw=FLEX_EXP_DIFF_RPT_SW-1;
//Configuring Diff Target Rate Calculation Period
MOD NRDUCELLUUPRFMOPT: NrDuCellId=0, DiffTargetRateCalcPeriod=2;
//Configuring the factor for estimating the differentiated target uplink rate
MOD NRDUCELLPUSCH: NrDuCellId=0,UlDiffTargetRateEstFactor=90;
//Configuring the factor for estimating the differentiated target downlink rate
MOD NRDUCELLUUPRFMOPT: NrDuCellId=0,DiffTargetRateEstFactor=90;

//Setting flexible experience differentiation configurations (The following uses three privilege levels as an
example, with rate limitation performed for (1) privilege level 0 only in the uplink based on the associated
target uplink data rate proportions and (2) the other privilege levels based on the associated target uplink
or downlink data rate proportions.)
ADD GNBFLEXEXPDIFFCFG: EquityLevel=0, UlTargetRateRatio=100, DlTargetRateRatio=100,
UlMinGteeBw=0, DlMinGteeBw=0,
EquityLevelAlgoSw=UL_UNLIMITED_RATE_SW-0&DL_UNLIMITED_RATE_SW-1;
ADD GNBFLEXEXPDIFFCFG: EquityLevel=1, UlTargetRateRatio=75, DlTargetRateRatio=75, UlMinGteeBw=0,
DlMinGteeBw=0, EquityLevelAlgoSw=UL_UNLIMITED_RATE_SW-0&DL_UNLIMITED_RATE_SW-0;
ADD GNBFLEXEXPDIFFCFG: EquityLevel=2, UlTargetRateRatio=50, DlTargetRateRatio=50, UlMinGteeBw=0,
DlMinGteeBw=0, EquityLevelAlgoSw=UL_UNLIMITED_RATE_SW-0&DL_UNLIMITED_RATE_SW-0;

//Configuring QCI-specific privilege levels (assuming that UEs with QCIs 6, 8, and 9 correspond to privilege
levels 0, 1, and 2 respectively)
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=6, EquityLevel=0;
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=8, EquityLevel=1;
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=9, EquityLevel=2;
```

```
//(Optional) Configuring the rate threshold for exiting flexible experience differentiation to avoid affecting the experience of UEs with poor channel conditions
MOD NRDUCELLRSVDEXT04: NrDuCellId=0, RsvdParam56=20;
MOD NRDUCELLPUSCH: NrDuCellId=0, UlFlexExpDiffExitRateThld=20;

//(Optional) It is recommended that CA Activate and BWP Switch UL Thld Slot Num and CA Activate and BWP Switch DL Thld Slot Num be set to 0 to improve the accuracy of data rate estimation when CA is enabled.
MOD NRDUCELLCARRMGMT:NrDuCellId=0, CaDlScellActThldSlotNum=0, CaDlScellDeactThldSlotNum=0;
MOD NRDUCELLCARRMGMT:NrDuCellId=0, CaUlScellActThldSlotNum=0, CaUlScellDeactThldSlotNum=0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch of data rate reporting in flexible experience differentiation
MOD NRDUCELLFEATURESW: NrDuCellId=0, QciBasedDiffServExpSw=FLEX_EXP_DIFF_RPT_SW-0;
//Turning off the flexible experience differentiation switch
MOD NRDUCELLFEATURESW: NrDuCellId=0, QciBasedDiffServExpSw=FLEX_EXP_DIFF_SW-0;
```

9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.4.2 Activation Verification

You can monitor the uplink or downlink throughput of a single UE to check whether this function has taken effect.

- When no minimum guaranteed bandwidth is configured, a lower target data rate proportion leads to a lower uplink or downlink throughput of users at the associated privilege level.
- In conditions of the same target data rate proportion, a higher minimum guaranteed bandwidth leads to a higher uplink or downlink throughput of users at the associated privilege level.

9.4.3 Network Monitoring

You can monitor whether this function has taken effect by comparing the average uplink and downlink throughputs of QCI-specific bearers for UEs in a cell before and after this function is enabled:

- Average downlink throughput of 5QI-specific bearers for UEs in a cell (at the RLC layer excluding tail-packet transmission) = 1000 x (**N.ThpVol.DL.DuCell5QI – N.ThpVol.DL.LastSlot.DuCell5QI**)/ **N.ThpTime.DL.RmvLastSlot.DuCell5QI**
- Average uplink throughput of 5QI-specific bearers for UEs in a cell (at the RLC layer excluding small-packet transmission) = 1000 x (**N.ThpVol.UL.DuCell5QI – N.ThpVol.UL.SmallPkt.DuCell5QI**)/ **N.ThpTime.UL.RmvSmallPkt.DuCell5QI**

- Average downlink throughput of QCI-specific bearers for UEs in a cell (at the RLC layer excluding tail-packet transmission) = 1000 x (**N.ThpVol.DL.QCI – N.ThpVol.DL.LastSlot.QCI**)/**N.ThpTime.DL.RmvLastSlot.QCI**
- Average uplink throughput of QCI-specific bearers for UEs in a cell (at the RLC layer excluding small-packet transmission) = 1000 x (**N.ThpVol.UL.QCI – N.ThpVol.UL.SmallPkt.QCI**)/**N.ThpTime.UL.RmvSmallPkt.QCI**

In addition, you can monitor the network impacts of this function by using the indicators in **9.2.2 Impacts**.

10 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

11 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

12 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

13 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- *Carrier Aggregation*
- *Modulation Schemes*
- *MIMO (TDD)*
- *VoNR*
- *ViNR*
- *Uplink Scheduling*
- *FWA*
- *DRX*
- *Channel Management*
- *QoS Management*
- *Power Control*
- *License Control Item Lists*
- *URLLC*
- *ANR*
- *PCI Conflict Detection and Self-Optimization*
- *Ultimate Cloud X Experience*
- *CA Experience Improvement*
- *UE Power Saving*
- *EPC-based NSA Experience Improvement*
- *Network Slicing*
- *Multi-Operator Sharing*
- *Cell Management*
- *Super Uplink*
- *RedCap*
- *Resource Control in NSA Networking*